On sliced Cramér metrics

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Abstract

This paper studies the family of sliced Cramér metrics, quantifying their stability under distortions of the input functions. Our results bound the growth of the sliced Cramér distance between a function and its geometric deformation by the product of the deformation's displacement size and the function's mean mixed norm. These results extend to sliced Cramér distances between tomographic projections. In addition, we remark on the effect of convolution on the sliced Cramér metrics. We also analyze efficient Fourier-based discretizations in 1D and 2D, and prove that they are robust to heteroscedastic noise. The results are illustrated by numerical experiments.

1 Introduction

This paper is concerned with properties of the sliced p-Cramér metrics, a family of distances that extend the classical univariate p-Cramér metrics to functions in $\mathbb{R}^d$ via the "slicing" operation. The Cramér and sliced Cramér metrics have arisen in a variety of different contexts. These include statistical testing [56, 71, 60, 72, 12] and probability theory [85]; machine learning applications, where they have been proposed as an alternative to Wasserstein and sliced Wasserstein distances [6, 40, 29]; and image processing [65]. The sliced 2-Cramér distance is also known as the energy distance [71, 60, 72, 54]. Well-known special cases in one variable include the 1-Cramér distance, which coincides with the 1-Wasserstein distance, and the ∞-Cramér distance, which coincides with the Kolmogorov metric.

A fundamental question about any metric is how stable it is under distortions of its inputs, such as geometric deformations or noise. It appears that, in spite of the long-standing interest in Cramér and sliced Cramér metrics, there has been little work on characterizing their stability. In this paper, we address this topic by describing stability properties of the sliced Cramér distances in settings of practical interest. We provide bounds on sliced Cramér distances' growth under geometric deformations of the inputs, and compare and contrast these bounds with those of Wasserstein and Lebesgue distances. We examine sliced Cramér metrics' behavior under convolutions, a setting which occurs frequently in signal and image processing applications. We also bound the sampling error of Fourier-based discretizations in 1D and 2D, and prove their robustness to heteroscedastic sub-Gaussian noise.

1.1 Geometric deformations

A natural question to ask about a metric is how stable it is to deformations of its inputs. For example, in image processing, one may seek a metric that is insensitive to small translations, rotations, changes of scale, or perturbations in the parameters that generate the images being compared. To make this question more precise, one must specify a class of deformations and a reasonable measure of the "size" of a deformation. There are many classes of deformations of practical interest, and in this paper we consider the family of push-forward deformations of a function: if Φ is a C^1 bijection mapping into the domain of a function f, it induces the deformation $f_\Phi(x) = f(\Phi(x))|\nabla\Phi(x)|$ of f. Such deformations preserve f's integral and L^1 norm. Other works have considered metric stability under different classes of deformations, such as those that preserve f's L^∞ norm [43, 3]; however, integral-preserving deformations are more natural for understanding sliced Cramér metrics, which are typically used to compare probability measures.

Given this class of deformations, a natural definition of a deformation Φ's size is its maximum displacement, $\max_x |x - \Phi(x)|$. In this paper, we will consider a family of measures of deformation size, defined in Section 2.1, which includes the maximum displacement as a special case. Proposition 2.1 in Section 2.1 relates these different measures of deformation size.

Having defined an appropriate measure of the size $\varepsilon(\Phi)$ of a deformation, we can quantify a metric D's robustness to deformations by the rate of growth of $\mathrm{D}(f, f_\Phi)$ as a function of $\varepsilon(\Phi)$. Because of the triangle inequality, this growth rate controls how "smoothly" the metric changes under deformations. More precisely, if D satisfies a bound of the form $\mathrm{D}(f, f_\Phi) \leq C(f)\varepsilon(\Phi)$, then the triangle inequality implies that for any functions f and g,

$$|\mathrm{D}(f, g) - \mathrm{D}(f_\Phi, g)| \leq \mathrm{D}(f, f_\Phi) \leq C(f)\varepsilon(\Phi). \tag{1}$$

This bound can be interpreted (see Remarks 1 and 10) as saying that $\mathrm{D}(f_\Phi, g)$ is a "smooth" function of the deformation Φ, when Φ is close to the identity transformation.

Prior work on metrics' behavior under deformations have focused primarily on Wasserstein-type distances [38, 58, 65]. Specifically, [58] and [65] show that they are robust to rigid deformations (translations and rotations) when comparing 2D tomographic projections of a 3D volume; these are useful properties for comparing images in single particle cryogenic electron microscopy (cryo-EM) [68, 7, 15], as well as other scientific problems for which the measurement modality only permits observing projections of an object [13, 49, 26, 27, 69, 11]. In Section 2 we state fairly general robustness properties of Wasserstein and sliced Wasserstein distances, which, though we have not seen them published previously, follow straightforwardly from prior work.

In this paper, we bound the growth rate of the sliced Cramér metrics under general deformations, and extend these bounds to the distances between tomographic projections. The growth rates depend both on the aforementioned measures of displacement size and on the function's mean mixed norm, which we define in Section 2.2. We also compare and contrast the bounds we prove for sliced Cramér metrics to the behavior of Wasserstein-type distances and Lebesgue distances.

1.2 Convolutions

In typical scientific applications, each observation will be a convolution of the underlying physical object with a filter that arises from the measurement apparatus. It is therefore of interest to understand how a metric changes when the inputs are each convolved by a common function. Since convolution is a smoothing operation, it is natural to expect that the convolved functions will be closer to each other (up to a scaling factor) than the clean functions. Indeed, this is the case for L^p distances (due to Young's convolution inequality), and is also true for Wasserstein and sliced Wasserstein distances (see Theorem 2.4 and Corollary 2.5 below). In this paper, we observe that the same is also true for sliced Cramér distances. This fact is essentially a rephrasing of a result from the prior work [80], where it is stated in terms of random variables (and specific to the case $p = 2$); we simply restate their result (generalized to all $p \geq 1$) in a way that is more directly applicable in the context of signal and image processing (and which also applies to signed functions, not just probablity distributions).

1.3 Discretizations and additive noise

Of course, for a metric to be of practical interest there must exist computationally efficient algorithms for computing it based on finite data. In this work we study the convergence of Fourier-based discretizations of the 1D Cramér metric and 2D sliced Cramér metric; the latter is based on the method described in [65], and may be computed rapidly via the non-uniform fast Fourier transform (NUFFT). These approximations take as inputs samples of the functions on an equispaced grid, as is standard in signal processing applications. We provide bounds on the discretization error when approximating certain classes of functions.

In addition to bounding the approximation errors on noiseless samples, we will also show that these discretizations are robust to additive, heteroscedastic, sub-Gaussian noise: that is, when applied to samples from a signal-plus-noise model, the estimated distance converges to the distance between the signals only, and filters out the noise. This is a useful property for a metric to have, though one that is by no means unique to the sliced Cramér metrics (there are many methods for filtering out noise). By contrast, many Wasserstein-type distances are not a priori defined (without modification) on noisy data, as they typically require their inputs to be probability measures; and Lebesgue distances do not possess such a robustness property, and can be severely distorted by additive noise.

1.4 Notation and conventions

Throughout the paper, we will assume familiarity with basic concepts of measure and integration, e.g. at the level of [19] or [25], and probability theory at the level of [75]. In this section we briefly review several definitions and introduce the notational conventions that we will use.

1.4.1 Lebesgue norms

Throughout, we denote by $\|f\|_{L^p} = \left(\int_{\mathbb{R}^d} |f(x)|^p dx\right)^{1/p}$ the Lebesgue p-norm of a function $f : \mathbb{R}^d \to \mathbb{R}$ (with the obvious modification when $p = \infty$), and denote the standard inner product as $\langle f, g\rangle = \int_{\mathbb{R}^d} f(x)g(x)dx$. If $A \subset \mathbb{R}^d$, we will say that f is in $L^p(A)$ if f is defined on $\mathbb{R}^d$ with $\|f\|_{L^p} < \infty$ and $f(x) = 0$ for $x \notin A$. We also define the p-norm $\|x\|_p$ for vectors x in $\mathbb{R}^d$ as $\|x\|_p = \left(\sum_{j=1}^d |x_j|^p\right)^{1/p}$ (again, with the obvious modification when $p = \infty$). When convenient, we will also denote the 2-norm of a vector x in $\mathbb{R}^d$ by $|x|$. We also denote the inner product between two vectors x and y in $\mathbb{R}^d$ by $\langle x, y\rangle = \sum_{j=1}^d x_j y_j$.

1.4.2 Absolute continuity

For a given interval $(a, b) \subset \mathbb{R}$, we denote by $\mathcal{A}_0 = \mathcal{A}_0(a, b)$ the set of absolutely continuous functions G on (a, b) satisfying $G(b) = 0$; note that such functions may be written in the form $G(x) = -\int_x^b g(t)\,dt$ where $g \in L^1$ and $g = G'$ almost everywhere.

1.4.3 The Fourier transform

We denote the Fourier transform of a function $f : \mathbb{R}^d \to \mathbb{R}$ in L^1 by $\widehat{f}(\xi) = \int_{\mathbb{R}^d} f(x)e^{-2\pi \mathrm{i}\langle x,\xi\rangle}dx$. With this convention, the Fourier inversion formula may be written as $f(x) = \int_{\mathbb{R}^d} \widehat{f}(\xi)e^{2\pi \mathrm{i}\langle x,\xi\rangle}d\xi$.

1.4.4 C^r functions

Let $r > 0$, and write $r = m + s$ where m is an integer and $0 \le s < 1$. Let $f : \mathbb{R}^d \to R$ be compactly supported. We say that f is C^r if all partial derivatives of f of degree up to and including m exist and are continuous, and, if $s > 0$, the partial derivatives of degree m are s-Hölder; that is, there is a $C > 0$ such that for all $x \in \mathbb{R}^d$ and $i = 1, \ldots, d$,

$$\sup_{h\neq 0} \frac{\partial^m f(x + he_i) - \partial^m f(x)}{|h|^s} \le C, \tag{2}$$

where $\partial^m f$ denotes any partial derivative of degree m. As is well-known, for such f, $|\widehat{f}(\xi)| = O(|\xi|^{-r})$.

1.4.5 Tomographic projections and the Radon transform

Let $\mathcal{U} = \left(u^{(1)}, \ldots, u^{(r)}\right) \in \mathbb{S}^{d-1} \times \cdots \times \mathbb{S}^{d-1}$ (where $\mathbb{S}^{d-1} \subset \mathbb{R}^d$ is the $(d-1)$-dimensional unit sphere) denote an ordered collection of r orthonormal vectors in $\mathbb{R}^d$. Let $u^{(r+1)}, \ldots, u^{(d)}$ denote $d - r$ orthonormal vectors that are orthogonal to $u_1, \ldots, u_r$. We define the tomographic projection $\mathcal{P}_{\mathcal{U}}$ onto $\mathrm{span}\{u^{(1)}, \ldots, u^{(r)}\}$ by

$$(\mathcal{P}_{\mathcal{U}} f)(t_1, \ldots, t_r) = \int_{\mathbb{R}^{d-r}} f(t_1u^{(1)} + \cdots + t_ru^{(r)} + s_1u^{(r+1)} + \cdots + s_{d-r}u^{(d)})\,ds. \tag{3}$$

When $r = 1$, we denote the tomographic projection of f onto the span of a unit vector u by $\mathcal{P}_u f$. Note that in this case, the *Radon transform* $\mathcal{R}f : \mathbb{R} \times \mathbb{S}^{d-1}$ of the function f is defined by $(\mathcal{R}f)(t, u) = (\mathcal{P}_u f)(t)$. For more background on these transforms, see, for example, the references [49, 26]. A standard result that we will use is the *Fourier slice theorem*: $\widehat{(\mathcal{P}_u f)}(\xi) = \widehat{f}(\xi u)$, for any unit vector $u \in \mathbb{R}^d$ and real number ξ.

1.4.6 Push-forwards

If $\Omega \subset \mathbb{R}^d$ is a (non-empty) open set, μ is a finite, signed measure on Ω, and $\Psi : \Omega \to \mathbb{R}^d$ is a measurable function, we denote by $\Psi_\sharp\mu$ the push-forward measure, $(\Psi_\sharp\mu)(E) = \mu(\Psi^{-1}(E))$; e.g. see [54]. Note that $(\Psi_\sharp\mu)(\Psi(\Omega)) = \mu(\Psi^{-1}(\Psi(\Omega))) = \mu(\Omega)$. When μ is induced from a function f supported on Ω, i.e. $\mu(E) = \int_E f(x)dx$, and Ψ is a diffeomorphism between Ω and $\Psi(\Omega)$ with inverse $\Phi = \Psi^{-1}$, then $\Psi_\sharp\mu$ has density $f(\Phi(x))|\det(\nabla\Phi(x))|$. We will write $(\Psi_\sharp f)(x) = (\Phi_\sharp^{-1} f)(x) = f(\Phi(x))|\det(\nabla\Phi(x))|$, or $f_\Phi(x) = f(\Phi(x))|\det(\nabla\Phi(x))|$ for short.

1.4.7 Sub-Gaussian random variables

If Z is a sub-Gaussian random variable, we denote by $\|Z\|_{\psi_2}$ its sub-Gaussian norm, defined by

$$\|Z\|_{\psi_2} = \inf\{\sigma > 0 \,:\, \mathbb{E}[\exp\{Z^2/\sigma^2\}] \le 2\}. \tag{4}$$

1.5 Outline of the remainder of the paper

The remainder of the paper is structured as follows:

1. Section 2 contains key definitions, background material, and several preliminary results. Theorem 2.2 and Theorem 2.3 establish general robustnes properties of Wasserstein and sliced Wasserstein distances, respectively; these results appear to be new, though they follow easily from prior literature.

2. Section 3 states the main theorems on the robustness of sliced Cramér distances to geometric deformations. Theorem 3.1 and Corollary 3.2 give bounds under quite general conditions. Sharper bounds are proved for several special cases of interest in Section 3.2, including rotations, translations, and dilations. In addition, Theorem 3.13 describes the behavior of sliced Cramér metrics under convolutions.

3. Section 4 analyzes Fourier-based approximations to the Cramér distances and the 2D sliced Cramér distances, the latter with respect to the uniform measure over $\mathbb{S}^1$. Theorems 4.1 and 4.6 show that these discretizations are robust to additive, heteroscedastic, sub-Gaussian noise.

4. Section 5 shows the results of several numerical experiments illustrating the theoretical results, including comparisons between the sliced Cramér, sliced Wasserstein, and Lebesgue distances.

5. Section 6 contains results on Fourier analysis, approximation theory, and concentration of measure that are used in the proofs from Sections 3 and 4.

6. Section 7 concludes the paper, providing a summary and topics for future research.

2 Background and preliminary results

This section introduces key definitions and background results that will be referred to in the rest of the paper. We draw particular attention to Theorem 2.2 and Theorem 2.3 on the growth of Wasserstein and sliced Wasserstein distances under deformations, which we have not seen published in the literature (but which follow straightforwardly from previously published results). These results will provide a useful point of reference for our findings on sliced Cramér metrics.

2.1 Deformations and displacement

Given an open, non-empty $\Omega \subset \mathbb{R}^d$ and a C^1, invertible Ψ defined on Ω with inverse $\Phi = \Psi^{-1}$, we will call the push-forward $f_\Phi(x) = f(\Phi(x))|\det(\nabla\Phi(x))|$ a *deformation of* f, and will also call the mappings Φ and Ψ themselves deformations.

2.1.1 Maximum displacement

We define the *maximum displacement* of the deformation Φ by

$$\varepsilon_\infty(\Phi) = \max_{x\in\Psi(\Omega)} |x - \Phi(x)|. \tag{5}$$

For example, if u is a fixed unit vector, then the function $\Psi(x) = x + \epsilon u$ has maximum displacement $\varepsilon_\infty(\Psi) = \epsilon$.

If u is a unit vector and Ψ is a C^1, 1-to-1 mapping defined on Ω, we define the maximum displacement of Ψ along u to be

$$\varepsilon(\Psi, u) = \max_{x\in\Omega} |\langle x - \Psi(x), u\rangle|. \tag{6}$$

Note that the maximum displacement can be written as

$$\varepsilon_\infty(\Psi) = \max_{u\in\mathbb{S}^{d-1}} \varepsilon(\Psi, u). \tag{7}$$

Remark 1. If we define the distance δ between deformations on a domain Ω by

$$\Delta(\Psi,\Phi) = \max_{y\in\Omega} |\Psi(y) - \Phi(y)|, \tag{8}$$

then one can write $\Delta(\Psi,\Phi)$ in terms of the maximum displacement ε_∞:

$$\Delta(\Phi,\Psi) = \varepsilon_\infty(\Phi \circ \Psi^{-1}). \tag{9}$$

Indeed,

$$\varepsilon_\infty(\Phi \circ \Psi^{-1}) = \max_{x\in\Psi(\Omega)} |x - \Phi(\Psi^{-1}(x))| = \max_{y\in\Omega} |\Psi(y) - \Phi(y)|. \tag{10}$$

2.1.2 Mean displacement

For a given probability distribution η over $\mathbb{S}^{d-1}$, and a value $1 \le p < \infty$, we define the *mean displacement* of Ψ as

$$\varepsilon_{\eta,p}(\Psi) = \left(\int_{\mathbb{S}^{d-1}} \varepsilon(\Psi,u)^p \, d\eta(u) \right)^{1/p}. \tag{11}$$

Note that if $\Phi = \Psi^{-1}$, then for all u,

$$\begin{aligned}
\varepsilon(\Psi,u) &= \max_{x\in\Omega} |\langle x - \Psi(x), u\rangle| \\
&= \max_{y\in\Psi(\Omega)} |\langle \Phi(y) - \Psi(\Phi(y)), u\rangle| \\
&= \max_{y\in\Psi(\Omega)} |\langle \Phi(y) - y, u\rangle| \\
&= \varepsilon(\Phi,u).
\end{aligned} \tag{12}$$

It then follows immediately that $\varepsilon_\infty(\Phi) = \varepsilon_\infty(\Psi)$ and $\varepsilon_{\eta,p}(\Phi) = \varepsilon_{\eta,p}(\Psi)$. When η is the uniform measure, we will denote $\varepsilon_{\eta,p}(\Psi)$ by $\varepsilon_p(\Psi)$. Also, when $d = 1$, all measures of distortion are identical, and we will denote their common value by $\varepsilon(\Psi)$.

Clearly, $\varepsilon_{\eta,p}(\Psi) \le \varepsilon_\infty(\Psi)$ for all p. The following result shows what is lost when the inequality is reversed:

Proposition 2.1. *Let $x^* = \arg\max_x |x - \Psi(x)|$, and let $u^* = (x^* - \Psi(x^*))/|x^* - \Psi(x^*)|$. Then*

$$\varepsilon_{\eta,p}(\Psi) \ge \varepsilon_\infty(\Psi) \cdot \left(\int_{S^{d-1}} |\langle u^*, u\rangle|^p \, d\eta(u) \right)^{1/p}. \tag{13}$$

Proof. By definition, $\varepsilon_\infty(\Psi) = |x^* - \Psi(x^*)|$. For all unit vectors u,

$$\varepsilon(\Psi,u) = \max_x |\langle x - \Psi(x), u\rangle| \ge |\langle x^* - \Psi(x^*), u\rangle| = |x^* - \Psi(x^*)| \cdot |\langle u^*, u\rangle| = \varepsilon_\infty(\Psi) \cdot |\langle u^*, u\rangle|. \tag{14}$$

The result then follows from averaging over u.

□

For example, if $d = 2$ and η is the uniform measure over $\mathbb{S}^1$,

$$\left(\int_{S^1} |\langle u^*, u\rangle|^p \, d\eta(u) \right)^{1/p} = \left(\frac{2}{\pi} \int_0^{\pi/2} \cos(\theta)^p \, d\theta \right)^{1/p} = \left(\frac{\Gamma(p/2 + 1/2)}{\Gamma(p/2+1)\sqrt{\pi}} \right)^{1/p}. \tag{15}$$

2.2 Mean mixed norms

Fix a probability measure η over the unit sphere $\mathbb{S}^{d-1}$ in $\mathbb{R}^d$. We define the *mean mixed norm* of $f : \mathbb{R}^d \to \mathbb{R}$ by

$$\|f\|_{M_\eta^{p,r}} = \left(\int_{\mathbb{S}^{d-1}} \|\mathcal{P}_u(|f|)\|_{L^p(\mathbb{R})}^r d\eta(u) \right)^{1/r} \tag{16}$$

for any $1 \le p \le \infty$ and $1 \le r < \infty$; and

$$\|f\|_{M^{p,\infty}} = \|f\|_{M_\eta^{p,\infty}} = \operatorname*{ess\,sup}_{u\in\mathbb{S}^{d-1}} \|\mathcal{P}_u(|f|)\|_{L^p(\mathbb{R})}. \tag{17}$$

When $p = r$, we will define $\|f\|_{M_\eta^p} \equiv \|f\|_{M_\eta^{p,p}}$.

Remark 2. For a fixed u, $\|\mathcal{P}_u(|f|)\|_{L^p(\mathbb{R})}$ is an example of a *mixed norm* [28], namely, the L^p norm of the L^1 norm of f. The mean mixed norm is then obtained by averaging this mixed norm over the choice of L^p variable.

Remark 3. If f is supported on a bounded open set and $f \in L^p$, then $\|f\|_{M_\eta^{p,r}} < \infty$. More precisely, if the support of f has diameter $L > 0$, then, $\|f\|_{M_\eta^{p,r}} \leq L^{(d-1)(p-1)/p}\|f\|_{L^p}$.

Remark 4. If η is the uniform measure, or if $r = \infty$, then $\|f\|_{M_\eta^{p,r}}$ is rotationally-invariant.

2.3 Sliced metrics

Given a metric $\mathrm{D}(f, g)$ between univariate functions, a value $p \geq 1$, and a probability density η over the unit sphere in $\mathbb{R}^d$, one can define a corresponding *sliced* metric $\mathrm{SD}_{\eta,p}(f, g)$ defined between functions f and g of d variables, as follows:

$$\mathrm{SD}_{\eta,p}(f, g) = \left(\int_{\mathbb{S}^{d-1}} \mathrm{D}(\mathcal{P}_u f, \mathcal{P}_u g)^p d\eta(u)\right)^{1/p}, \tag{18}$$

with the obvious modification when $p = \infty$. That is, $\mathrm{SD}_{\eta,p}(f, g)$ is obtained by averaging the distances between the one-dimensional projections of f and g over all directions. For background on sliced metrics, see, for instance, the references [55, 9, 31, 32, 14, 52, 65, 50, 51]. In typical applications of sliced metrics, η is the uniform measure over $\mathbb{S}^{d-1}$; however, other choices and methods for choosing η have been proposed in the literature [50, 51], and in this paper we consider general η. There also exist generalizations of sliced metrics to other geometries [34, 74], though we will not consider these in the present work.

2.4 Wasserstein distances

Recall that if f and g are probability densities on a subset $\Omega \subset \mathbb{R}^d$, their *$p$-Wasserstein distance* $\mathrm{W}_p(f, g)$ (also known as the Kantorovich distance) is defined as

$$\mathrm{W}_p(f, g) = \inf_{\Pi \in \mathcal{M}(f,g)} \left(\int_\Omega \int_\Omega |x - y|^p d\Pi(x, y)\right)^{1/p}, \tag{19}$$

where $\mathcal{M}(f, g)$ denotes the space of all probability measures on $\Omega \times \Omega$ with marginals equal to f and g, respectively [76, 77]. That is, $\Pi \in \mathcal{M}(f, g)$ if for all measurable $E \subset \Omega$,

$$\Pi(E \times \Omega) = \int_E f(x)dx, \tag{20}$$

and

$$\Pi(\Omega \times E) = \int_E g(y)dy. \tag{21}$$

Informally, $\mathrm{W}_p(f, g)$ is the minimal cost of rearranging a unit of mass with distribution f into one with distribution G, where the cost of moving mass between locations x and y is $|x - y|^p$. The distance $\mathrm{W}_1(f, g)$ is also known as the *Earth Mover's Distance (EMD)* between the probability measures f and g [76, 77]. The Wasserstein distances and their variants have been widely used in statistics, machine learning, image processing, and related areas [53, 54, 63, 9, 55, 62, 39, 8, 59, 47, 10].

2.4.1 Wasserstein distances and deformations

The Wasserstein distance is a relaxation of the Monge distance, defined by

$$\mathrm{M}_p(f, g) = \inf_{\Phi \in \mathcal{T}(f,g)} \left(\int_\Omega |x - \Phi(x)|^p f(x)\, dx\right)^{1/p} \tag{22}$$

where $\mathcal{T}(f, g)$ contains those functions $\Phi : \Omega \to \Omega$ such that $\int_E g(x)dx = \int_{\Phi^{-1}(E)} f(x)dx$, that is, which push f onto g. Indeed, any Φ in $\mathcal{T}(f, g)$ induces a measure Π_Φ in $\mathcal{M}(f, g)$, with

$$\int_\Omega \int_\Omega |x - y|^p d\Pi_\Phi(x, y) = \int_\Omega |x - \Phi(x)|^p f(x)\, dx, \tag{23}$$

and hence $\mathrm{W}_p(f,g) \le \mathrm{M}_p(f,g)$. (In fact, when $\mathrm{M}_p(f,g)$ is finite, equality holds; see [63].) Consequently, if Φ is a smooth bijection on Ω and $f_\Phi(x) = f(\Phi(x))|\det(\nabla\Phi(x))|$, then Φ is contained in $\mathcal{T}(f, f_\Phi)$, and so

$$\mathrm{W}_p(f, f_\Phi) \le \mathrm{M}_p(f, f_\Phi) \le \left(\int_\Omega |x - \Phi(x)|^p f(x)\,dx\right)^{1/p} \le \varepsilon_\infty(\Phi)\left(\int_\Omega f(x)\,dx\right)^{1/p} = \varepsilon_\infty(\Phi). \tag{24}$$

In fact, a more general robustness result may be easily shown, which we state now.

Theorem 2.2. *Suppose f is a probability density supported on a bounded, open set $\Omega \subset \mathbb{R}^D$, and let $\Phi : \Omega \to \Omega$ be a smooth bijection. Let $\mathcal{Q}$ be a tomographic projection operator onto a d-dimensional subspace, $d \le D$. Then for all $p \ge 1$,*

$$\mathrm{W}_p(\mathcal{Q}f, \mathcal{Q}f_\Phi) \le \varepsilon_\infty(\Phi). \tag{25}$$

Proof. An identical proof to that of Lemma 1 in [58] shows that $\mathrm{W}_p(\mathcal{Q}f, \mathcal{Q}f_\Phi) \le \mathrm{W}_p(f, f_\Phi)$ (note that the left side refers to transportation in $\mathbb{R}^d$, and the right side to $\mathbb{R}^D$). The result then follows from (24).

□

2.4.2 Wasserstein distances in 1D

It is well-known that Wasserstein distances in 1D take a on a particularly simple form. Denote by $\mathcal{V}$ the Volterra operator [23] on $L^1([a,b])$, defined by

$$(\mathcal{V}f)(x) = \int_a^x f(t)dt. \tag{26}$$

Then when $d = 1$, it is known [63] that $\mathrm{W}_p(f,g)$ may be written as follows:

$$\mathrm{W}_p(f,g) = \|(\mathcal{V}f)^{-1} - (\mathcal{V}g)^{-1}\|_{L^p}. \tag{27}$$

Here, $(\mathcal{V}f)^{-1}$ denotes the functional inverse of $\mathcal{V}f$, defined as

$$(\mathcal{V}f)^{-1}(x) = \inf\{t \in [a,b] \ : \ (\mathcal{V}f)(t) \ge x\}. \tag{28}$$

When $p = 1$, it is also true that $\mathrm{W}_1(f,g) = \|\mathcal{V}f - \mathcal{V}g\|_{L^1}$.

2.4.3 Sliced Wasserstein

Given $p \ge 1$ and a probability density η over $\mathbb{S}^{d-1}$, we denote by $\mathrm{SW}_{\eta,p}$ the sliced Wasserstein distance:

$$\mathrm{SW}_{\eta,p}(f,g) = \left(\int_{\mathbb{S}^{d-1}} \mathrm{W}_p(\mathcal{P}_u f, \mathcal{P}_u g)^p d\eta(u)\right)^{1/p}. \tag{29}$$

Sliced Wasserstein distances have been the subject of considerable research activity in recent years [55, 9, 31, 32, 14, 52, 65]. The work [65] proves that sliced Wasserstein distances are robust to rotations and translations, and also describes a fast discretization.

We prove an analogue of Theorem 2.2 for sliced Wasserstein distances:

Theorem 2.3. *Suppose f is a probability density supported on a bounded, open set $\Omega \subset \mathbb{R}^D$, and let $\Phi : \Omega \to \Omega$ be a smooth bijection. Let $\mathcal{Q}$ be a tomographic projection operator onto d-dimensional subspace, $d \le D$. Then for all $p \ge 1$,*

$$\mathrm{SW}_{\eta,p}(\mathcal{Q}f, \mathcal{Q}f_\Phi) \le \varepsilon_\infty(\Phi). \tag{30}$$

Proof. Without loss of generality, suppose $\mathcal{Q}$ projects onto the first d coordinates. Let $u \in \mathbb{S}^{d-1}$. It is easy to see (and will be shown later, in the proof of Corollary 3.2 in Section 3) that $(\mathcal{P}_u\mathcal{Q}f)(t) = (\mathcal{P}_{(u,0)}f)(t)$, which is the tomographic projection of f onto the span of $(u,0) \in \mathbb{R}^d \times \mathbb{R}^{D-d}$. Consequently, from Theorem 2.2,

$$\mathrm{W}_p(\mathcal{P}_u\mathcal{Q}f, \mathcal{P}_u\mathcal{Q}f_\Phi) = \mathrm{W}_p(\mathcal{P}_{(u,0)}f, \mathcal{P}_{(u,0)}f_\Phi) \le \varepsilon_\infty(\Phi). \tag{31}$$

The result now follows by averaging over u.

□

2.4.4 Wasserstein distances and convolution

In signal and image processing applications, one typically observes signals/images that have been convolved with a filter induced from the measurement process. It is therefore of interest to understand how metrics behave when their inputs are convolved by a common function. For Wasserstein distances, we have the following result:

Theorem 2.4. *Suppose f, g and w are probability densities on $\mathbb{R}^d$. Then for all $p \geq 1$,*

$$\mathrm{W}_p(f * w, g * w) \leq \mathrm{W}_p(f, g). \tag{32}$$

This property is referred to by Zolotarev as *regularity* of the metric [84, 83, 85].[1] While the result is certainly known and is straightforward to prove, we had difficulty locating a reference containing a proof of the precise statement; for the reader's convenience we provide a proof here.

Proof of Theorem 2.4. By Kantorovich duality (see, e.g., Chapter 5 in [77]),

$$\mathrm{W}_p(f, g)^p = \sup_{(\varphi,\psi)\in\mathcal{F}_p} \left\{ \int f(x)\varphi(x)\, dx + \int g(y)\psi(y)\, dy \right\}, \tag{33}$$

where $\mathcal{F}_p$ contains all pairs (φ, ψ) of integrable functions φ and ψ satisfying $\varphi(x) + \psi(y) \leq |x - y|^p$. Take any such φ and ψ. Letting $\widetilde{w}(z) = w(-z)$, we have

$$\int (f * w)(x)\varphi(x)\, dx = \int (\varphi * \widetilde{w})(z) f(z)\, dz, \tag{34}$$

and

$$\int (g * w)(y)\psi(y)\, dy = \int (\psi * \widetilde{w})(z) g(z)\, dz. \tag{35}$$

For any x and y we have

$$(\varphi * \widetilde{w})(x) + (\psi * \widetilde{w})(y) = \int (\varphi(x - z) + \psi(y - z))\widetilde{w}(z)\, dz \leq |x - y|^p \int \widetilde{w}(z)\, dz = |x - y|^p. \tag{36}$$

Therefore, $(\varphi * \widetilde{w}, \psi * \widetilde{w}) \in \mathcal{F}_p$, and

$$\int (f * w)(x)\varphi(x)\, dx + \int (g * w)(y)\psi(y)\, dy = \int (\varphi * \widetilde{w})(z) f(z)\, dz + \int (\psi * \widetilde{w})(z) g(z)\, dz \leq \mathrm{W}_p(f, g), \tag{37}$$

and so taking the supremum over all $(\varphi, \psi) \in \mathcal{F}_p$ proves the result.

□

Corollary 2.5. *Suppose f, g and w are probability densities on $\mathbb{R}^d$, and let η be a probability density over $\mathbb{S}^{d-1}$. Then for all $p \geq 1$,*

$$\mathrm{SW}_p(f * w, g * w) \leq \mathrm{SW}_p(f, g). \tag{38}$$

Proof. It is straightforward to see (and will be shown in Section 3.3) that for any unit vector u in $\mathbb{R}^d$, $\mathcal{P}_u(f * w) = (\mathcal{P}_u f) * (\mathcal{P}_u w)$. Then from the $d = 1$ case of Theorem 2.4,

$$\mathrm{W}_p(\mathcal{P}_u(f * w), \mathcal{P}_u(g * w)) = \mathrm{W}_p((\mathcal{P}_u f) * (\mathcal{P}_u w), (\mathcal{P}_u g) * (\mathcal{P}_u w)) \leq \mathrm{W}_p(\mathcal{P}_u f, \mathcal{P}_u g). \tag{39}$$

Averaging over all $u \in \mathbb{S}^{d-1}$ then proves the result.

□

Remark 5. Young's convolutional inequality (e.g. see Chapter 8 in [19]) states that the same property holds for the ordinary L^p distances on $\mathbb{R}^d$, namely, for f and g in L^p and w in L^1, $\|f * w - g * w\|_{L^p} \leq \|w\|_{L^1}\|f - g\|_{L^p}$.

[1] An equivalent statement of regularity for a metric D between random variables is that for all random variables X, Y and Z, where Z is independent of X and Y, $D(X + Z, Y + Z) \leq D(X, Y)$.

2.5 Cramér and sliced Cramér metrics

We introduce the primary objects of interest in this paper: the Cramér and sliced Cramér distances.

2.5.1 Cramér distances

Let f be in $L^1(a,b)$. For any value $1 \leq p \leq \infty$, we will refer to $\|f\|_{V^p} \equiv \|\mathcal{V}f\|_{L^p}$ as the *Volterra p-norm* of f. Note that, because $\mathcal{V}f$ is in $L^\infty(a,b)$, the Volterra p-norm of f is finite for any function f in $L^1(a,b)$. We denote by $\mathrm{C}_p(f,g) = \|f-g\|_{V^p}$ the *p-Cramér metric* (or *p-Cramér distance*) between functions f and g, named after Harald Cramér [12, 60, 71].[2]

Remark 6. While the Cramér metrics are typically used to compare probability distributions, they are well-defined for any f and g in $L^1(a,b)$. Note, however, that it is most natural to compare functions supported on (a,b) and for which $\int_a^b f = \int_a^b g$, since otherwise the distance may change by enlarging the interval. The results in this paper assume that $\int_a^b f = \int_a^b g$; however, they do not require f and g to be non-negative.

The following result provides a dual formulation of the Volterra p-norm (and hence of the p-Cramér metric) that will be useful in our subsequent analysis. It essentially appears as Theorem 1 in [42]; because of its key role in this paper, we provide a self-contained proof for the reader's convenience.

Proposition 2.6. *Let $1 \leq p \leq \infty$ and let q be the conjugate exponent: $1/p + 1/q = 1$. Then for any function f in $L^1(a,b)$,*

$$\|f\|_{V^p} = \sup_{G \in \mathcal{A}_0 : \|G'\|_{L^q} \leq 1} \langle f, G \rangle. \tag{40}$$

Proof. First, note that the adjoint transform $\mathcal{V}^*$ is given by

$$(\mathcal{V}^* f)(x) = \int_x^b f(t)dt. \tag{41}$$

This operator satisfies

$$\langle \mathcal{V}f, g \rangle = \langle f, \mathcal{V}^* g \rangle \tag{42}$$

where f and g are two functions in $L^1(a,b)$.

By duality of L^p and L^q, we have:

$$\|f\|_{V^p} = \|\mathcal{V}f\|_{L^p} = \sup_{g : \|g\|_{L^q} \leq 1} \int_a^b (\mathcal{V}f)(x) g(x) dx = \sup_{g : \|g\|_{L^q} \leq 1} \langle \mathcal{V}f, g \rangle = \sup_{g : \|g\|_{L^q} \leq 1} \langle f, \mathcal{V}^* g \rangle. \tag{43}$$

Any function of the form $\mathcal{V}^* g$ is contained in $\mathcal{A}_0$, and any function G in $\mathcal{A}_0$ is of the form $G = \mathcal{V}^* g$ where $g = G'$ almost everywhere. Consequently:

$$\|f\|_{V^p} = \sup_{g : \|g\|_{L^q} \leq 1} \langle f, \mathcal{V}^* g \rangle = \sup_{G \in \mathcal{A}_0 : \|G'\|_{L^q} \leq 1} \langle f, G \rangle, \tag{44}$$

which completes the proof.

□

2.5.2 Sliced Cramér distances

Following the framework from Section 2.3, given a probability measure η over the unit sphere $\mathbb{S}^{d-1} \subset \mathbb{R}^d$, for all $1 \leq p < \infty$ we define *sliced p-Cramér metric* between $f, g : \mathbb{R}^d \to \mathbb{R}$ as

$$\mathrm{SC}_{\eta,p}(f,g) = \left(\int_{\mathbb{S}^{d-1}} \mathrm{C}_p(\mathcal{P}_u f, \mathcal{P}_u g)^p \, d\eta(u) \right)^{1/p}, \tag{45}$$

and when $p = \infty$ we define

$$\mathrm{SC}_\infty(f,g) = \mathrm{SC}_{\eta,\infty}(f,g) = \sup_{u \in \mathbb{S}^{d-1}} \mathrm{C}_\infty(\mathcal{P}_u f, \mathcal{P}_u g). \tag{46}$$

[2] In [85], Zolotarev refers to the p-Cramér distance more simply as the "L_p-metric".

Remark 7. Cramér and sliced Cramér metrics have garnered attention in recent years as metrics for comparing probability measures in machine learning applications [48, 30, 6, 40, 29] and image processing [65]. The ∞-Cramér metric is also known as the Kolmogorov Metric between f and g [22], and arises in the context of goodness-of-fit testing in statistics [21]. The 1-Cramér metric is equal to the 1-Wasserstein distance between the probability distributions f and g, described in Section 2.4. The sliced 2-Cramér distances are also equal (up to a change in scaling) to the multi-dimensional energy distances used in statistical testing [60, 71], as remarked upon in [78].

For a function f of two variables, we define its *sliced Volterra norm* by

$$\|f\|_{SV_\eta^p} = \left(\int_{\mathbb{S}^{d-1}} \|\mathcal{P}_u f\|_{L^p}^p \, d\eta(u) \right)^{1/p} \tag{47}$$

when $1 \leq p < \infty$, and

$$\|f\|_{SV^\infty} = \|f\|_{SV_\eta^\infty} = \sup_{u \in \mathbb{S}^{d-1}} \|\mathcal{P}_u f\|_{L^\infty}. \tag{48}$$

Then for two functions f and g, $\mathrm{SC}_{\eta,p}(f,g) = \|f-g\|_{SV_\eta^p}$. When η is the uniform measure over $\mathbb{S}^{d-1}$, we will denote the sliced Cramér metric more simply as $\mathrm{SC}_p(f,g)$, and the sliced Volterra norm more simply as $\|f\|_{SV^p}$.

In Section 3, we will study the geometric properties of sliced Cramér metrics, characterizing their growth under deformations. In Section 4 we will analyze efficient, Fourier-based discretizations for the 1D and 2D distances (the latter based on those in [65]) between functions with equal integrals, and prove their robustness to additive heteroscedastic noise.

3 Properties of sliced Cramér metrics

In this section, we will provide bounds on the sliced Cramér distance between a function and its deformation in terms of the deformation size. As described in Remark 10 below, these bounds also imply that the sliced Cramér distances are locally smooth with respect to the distance Δ between deformations defined in Section 2.1. We will also extend these results to the sliced Cramér distance between tomographic projections. We will compare these bounds to those satisfied by the Wasserstein and sliced Wasserstein distances found in Theorem 2.2 and Theorem 2.3, respectively, as well as to the Lebesgue distances. Bounds for general deformations are provided in Theorem 3.1 and Corollary 3.2, in Section 3.1. Sharper bounds are then derived for more specific deformations in Section 3.2. In addition, Section 3.3 analyzes the behavior of the sliced Cramér distances under convolution of the inputs, stating a bound analagous to Theorem 2.4 and Corollary 2.5.

3.1 Growth under deformations

We start with a general result that quantifies sliced Cramér metrics' growth under deformations.

Theorem 3.1. *Let $1 \leq p \leq \infty$. Suppose that A and B are non-empty, bounded, open sets in $\mathbb{R}^d$. Let $\Phi : B \to A$ be a C^1 bijection, and η be a probability measure over $\mathbb{S}^{d-1}$. If f is in $L^p(A)$, then*

$$\mathrm{SC}_{\eta,p}(f, f_\Phi) \leq 2^{(p-1)/p} \cdot \|f\|_{M_\eta^p} \cdot \varepsilon_\infty(\Phi) \tag{49}$$

and

$$\mathrm{SC}_{\eta,p}(f, f_\Phi) \leq 2^{(p-1)/p} \cdot \|f\|_{M^{p,\infty}} \cdot \varepsilon_{\eta,p}(\Phi). \tag{50}$$

If f is in $L^1(A)$, then

$$\mathrm{SC}_{\eta,p}(f, f_\Phi) \leq \|f\|_{L^1} \cdot \varepsilon_{\eta,1}(\Phi)^{1/p}. \tag{51}$$

We record several observations about Theorem 3.1, before stating a corollary and providing the proof.

Remark 8. Because $f = (f_\Phi)_{\Phi^{-1}}$ and $\varepsilon_{\eta,p}(\Phi) = \varepsilon_{\eta,p}(\Phi^{-1})$, one can switch the roles of f and f_Φ, and thereby replace the term $\|f\|_{M_\eta^p}$ by $\min\{\|f\|_{M_\eta^p}, \|f_\Phi\|_{M_\eta^p}\}$ in (49), and replace $\|f\|_{M_\eta^{p,\infty}}$ by $\min\{\|f\|_{M_\eta^{p,\infty}}, \|f_\Phi\|_{M_\eta^{p,\infty}}\}$ in (50), to obtain sharper bounds.

Remark 9. It is not possible to replace the mean mixed norms with the L^1 norm in (49) and (50). Indeed, when $d = 1$, let f be the point mass at 0, and let $\Phi(x) = x + \epsilon$; then $\mathrm{C}_p(f, f_\Phi) = \epsilon^{1/p}$, showing that a bound of the form $\mathrm{C}_p(f, f_\Phi) \le K(p)\|f\|_{L^1}\varepsilon(\Phi)$ is not possible in general.

Remark 10. One may reinterpret Theorem 3.1 as saying that, for fixed functions f and g, the mapping $\Phi \mapsto \mathrm{SC}(f_\Phi, g)$ is locally Lipschitz, with respect to the metric Δ defined in Remark 1, in a neighborhood of the identity. Indeed, defining $\mathrm{I}(x) = x$, the triangle inequality implies

$$|\mathrm{SC}_{\eta,p}(f_\Phi, g) - \mathrm{SC}_{\eta,p}(f, g)| \le \mathrm{SC}_{\eta,p}(f_\Phi, f) \le 2^{(p-1)/p} \cdot \|f\|_{M^p_\eta} \cdot \varepsilon_\infty(\Phi \circ \mathrm{I}^{-1}) = 2^{(p-1)/p} \cdot \|f\|_{M^p_\eta} \cdot \Delta(\Phi, \mathrm{I}). \tag{52}$$

Similarly, when f is in L^1 the distances are locally $1/p$-Hölder:

$$|\mathrm{SC}_{\eta,p}(f_\Phi, g) - \mathrm{SC}_{\eta,p}(f, g)| = \|f\|_{L^1} \cdot \Delta(\Phi, \mathrm{I})^{1/p}. \tag{53}$$

Remark 11. We compare the bounds from Theorem 3.1 to the bounds on the Wasserstein and sliced Wasserstein distances described in Sections 2.4 and 2.3. First, it is convenient to combine the bounds (49) and (51) to get the slightly weaker but simpler bound

$$\mathrm{SC}(f, f_\Phi) \le \begin{cases} 2^{(p-1)/p} \cdot \|f\|_{M^p_\eta} \cdot \varepsilon_\infty(\Phi), & \text{if } \varepsilon_\infty(\Phi) \le \frac{1}{2}\left(\frac{\|f\|_{L^1}}{\|f\|_{M^p_\eta}}\right)^{p/(p-1)} \\ \|f\|_{L^1} \cdot \varepsilon_\infty(\Phi)^{1/p}, & \text{otherwise} \end{cases}. \tag{54}$$

Like Wasserstein, $\mathrm{SC}_{\eta,p}(f, f_\Phi)$ is bounded by a linear function of the maximal displacement $\varepsilon_\infty(\Phi)$. However, we note two differences between the metrics. First, the sliced Cramér bound exhibits slower growth with respect to $\varepsilon_\infty(\Phi)$ when $\varepsilon_\infty(\Phi)$ is large, growing like $\varepsilon_\infty(\Phi)^{1/p}$ instead of $\varepsilon_\infty(\Phi)$. Second, for small $\varepsilon_\infty(\Phi)$, while both bounds are linear in $\epsilon_\infty(\Phi)$, they differ in their dependence on f, with the bound on the Wasserstein distance scaling with $\|f\|_{L^1}$ and the bound on the sliced Cramér distance scaling with $\|f\|_{M^p_\eta}$. While the norms $\|f\|_{L^1}$ and $\|f\|_{M^p_\eta}$ are generally not commensurable, we observe that if the support of f has diameter $\delta > 0$, say, then $\|f\|_{L^1} \le \delta^{(p-1)/p}\|f\|_{M^p_\eta}$; consequently, if f is a probability density, then

$$\mathrm{W}_p(f, f_\Phi) \le \varepsilon_\infty(\Phi) \le \delta^{(p-1)/p} \cdot \|f\|_{M^p_\eta} \cdot \varepsilon_\infty(\Phi), \tag{55}$$

which is smaller than the bound on $\mathrm{SC}_{\eta,p}(f, f_\Phi)$ when $\delta < 2$.

Remark 12. We compare Theorem 3.1 with the L^p distance, in the case of translations. Suppose f is supported on a set of diameter δ, with $\|f\|_{L^p} = 1$, and let $f_\epsilon(x) = f(x - \epsilon u)$, where u is a fixed unit vector. Then for any $\epsilon > \delta$, $\|f - f_\epsilon\|_{L^p} = 2^{1/p}$. Therefore, close to $\epsilon = 0$, viewing $\|f - f_\epsilon\|_{L^p}$ as a function of ϵ, its slope over $[0, \delta]$ is $2^{1/p}/\delta$, which can be arbitrarily big when δ is small, i.e. when f has a large oscillation. We contrast this with the behavior of $\mathrm{SC}_{\eta,p}(f, f_\Phi)$. Note that in the case of translations, Theorem 3.10 below shows that the factor $2^{(p-1)/p}$ in the bounds (49) and (50) may be removed, and so the sliced Cramér distances satisfy the simplified bound

$$\mathrm{SC}(f, f_\Phi) \le \begin{cases} \|f\|_{M^p_\eta} \cdot \epsilon, & \text{if } \epsilon \le \left(\frac{\|f\|_{L^1}}{\|f\|_{M^p_\eta}}\right)^{p/(p-1)} \\ \|f\|_{L^1} \cdot \epsilon^{1/p}, & \text{otherwise} \end{cases}. \tag{56}$$

With this, we make two observations. First, since $\|f\|_{L^1} \le \delta^{(p-1)/p} \cdot \|f\|_{M^p_\eta}$, the transition between the ϵ and $\epsilon^{1/p}$ regimes in (56) occurs at the translation size

$$\epsilon = \left(\frac{\|f\|_{L^1}}{\|f\|_{M^p_\eta}}\right)^{p/(p-1)} \le \delta; \tag{57}$$

this is qualitatively similar behavior to the Lebesgue distance, which also changes behavior when $\epsilon > \delta$, becoming constant. However, the sliced Cramér distance exhibits much slower (and hence smoother) growth than the Lebesgue distance in the small ϵ regime; indeed, we can bound the sliced Cramér distance, for f with $\|f\|_{L^p} = 1$, by

$$\mathrm{SC}_{\eta,p}(f, f_\epsilon) \le \|f\|_{M^p_\eta} \cdot \epsilon \le \delta^{(d-1)(p-1)/p} \cdot \|f\|_{L^p} \cdot \epsilon = \delta^{(d-1)(p-1)/p} \cdot \epsilon, \tag{58}$$

i.e. the slope is bounded by $\delta^{(d-1)(p-1)/p}$, in contrast to the Lebesgue distance's slope of $2^{1/p}/\delta$.

Theorem 3.1 is easily extended to comparing tomographic projections of a function and its deformation. This is of interest when measuring the distance between two 2D projections of a 3D volume, such as in the analysis of images in cryo-electron microscopy.

Corollary 3.2. *Let $1 \le p \le \infty$. Let A and B be non-empty, bounded, open sets in $\mathbb{R}^D$, f be in $L^p(A)$, $\Phi : B \to A$ be a C^1 bijection, and $f_\Phi(x) = f(\Phi(x))|\det(\nabla\Phi(x))|$ on B, and 0 elsewhere. For $d \le D$, let $\mathcal{Q}$ denote the tomographic projection operator onto a d-dimensional subspace of $\mathbb{R}^D$. Then for any probability measure η over $\mathbb{S}^{d-1}$,*

$$\mathrm{SC}_{\eta,p}(\mathcal{Q}f, \mathcal{Q}f_\Phi) \le 2^{(p-1)/p} \cdot \|\mathcal{Q}(|f|)\|_{M^p_\eta} \cdot \varepsilon_\infty(\Phi), \tag{59}$$

and

$$\mathrm{SC}_{\eta,p}(\mathcal{Q}f, \mathcal{Q}f_\Phi) \le \|f\|_{L^1} \cdot \varepsilon_\infty(\Phi)^{1/p}. \tag{60}$$

Proof of Corollary 3.2. Without loss of generality, suppose $\mathcal{Q}$ projects onto the first d coordinates, that is,

$$(\mathcal{Q}f)(x) = \int_{\mathbb{R}^{D-d}} f(x, y)dy. \tag{61}$$

Let u be a unit vector in $\mathbb{R}^d$, and let $u^{(2)}, \ldots, u^{(d)}$ denote any orthonormal vectors completing the basis, so that for h on $\mathbb{R}^d$,

$$(\mathcal{P}_u h)(t) = \int_{\mathbb{R}^{d-1}} h(tu + s_2u^{(2)} + \cdots + s_du^{(d)})\, ds, \tag{62}$$

and so

$$\begin{aligned}(\mathcal{P}_u(\mathcal{Q}h))(t) &= \int_{\mathbb{R}^{d-1}} (\mathcal{Q}h)(tu + s_2u^{(2)} + \cdots + s_du^{(d)})\, ds \\ &= \int_{\mathbb{R}^{d-1}} \int_{\mathbb{R}^{D-d}} h(tu + s_2u^{(2)} + \cdots + s_du^{(d)}, y)\, dy\, ds \\ &= (\mathcal{P}_{(u,0)}h)(t),\end{aligned} \tag{63}$$

which is the tomographic projection of h onto the span of $(u, 0) \in \mathbb{R}^d \times \mathbb{R}^{D-d}$.

Denote by $\widetilde{\eta}$ the distribution over the unit sphere $\mathbb{S}^{D-1}$, supported on $\widetilde{\mathbb{S}}^{d-1} \equiv \{(u, 0) \in \mathbb{R}^d \times \mathbb{R}^{D-d} : |u| = 1\}$ and defined by $d\widetilde{\eta}((u, 0)) = d\eta(u)$ for $u \in \mathbb{R}^d$. We have

$$\begin{aligned}\mathrm{SC}_{\widetilde{\eta},p}(f, f_\Phi)^p &= \int_{\mathbb{S}^{D-1}} \mathrm{C}_p(\mathcal{P}_v f, \mathcal{P}_v f_\Phi)^p\, d\widetilde{\eta}(v) \\ &= \int_{\mathbb{S}^{d-1}} \mathrm{C}_p(\mathcal{P}_{(u,0)} f, \mathcal{P}_{(u,0)} f_\Phi)^p\, d\eta(u) \\ &= \int_{\mathbb{S}^{d-1}} \mathrm{C}_p(\mathcal{P}_u \mathcal{Q}f, \mathcal{P}_u \mathcal{Q}f_\Phi)^p\, d\eta(u) \\ &= \mathrm{SC}_{\eta,p}(\mathcal{Q}f, \mathcal{Q}f_\Phi)^p.\end{aligned} \tag{64}$$

Furthermore,

$$\begin{aligned}\|f\|^p_{M^p_{\widetilde{\eta}}} &= \int_{\mathbb{S}^{D-1}} \|\mathcal{P}_v(|f|)\|^p_{L^p}\, d\widetilde{\eta}(v) \\ &= \int_{\mathbb{S}^{d-1}} \|\mathcal{P}_{(u,0)}(|f|)\|^p_{L^p}\, d\eta(u) \\ &= \int_{\mathbb{S}^{d-1}} \|\mathcal{P}_u \mathcal{Q}(|f|)\|^p_{L^p}\, d\eta(u) \\ &= \|\mathcal{Q}(|f|)\|^p_{M^p_\eta}.\end{aligned} \tag{65}$$

The inequality (59) then follows by applying (49) in Theorem 3.1 with the measure $\widetilde{\eta}$. The bound (60) follows from (51) and the fact that $\varepsilon_{\widetilde{\eta},1}(\Phi) \le \varepsilon_\infty(\Phi)$.

$\square$

The proof of Theorem 3.1 is immediate from the following lemma:

Lemma 3.3. *Let $1 \le p \le \infty$. Let A and B be non-empty, bounded, open sets in $\mathbb{R}^d$, f be in $L^p(A)$, $\Phi : B \to A$ be a C^1 bijection and $f_\Phi(x) = f(\Phi(x))|\det(\nabla\Phi(x))|$ on B, and 0 elsewhere. Then for any $u \in \mathbb{S}^{d-1}$,*

$$\mathrm{C}_p(\mathcal{P}_u f, \mathcal{P}_u f_\Phi) \le \min\left\{2^{(p-1)/p} \cdot \|\mathcal{P}_u(|f|)\|_{L^p} \cdot \varepsilon(\Phi, u),\ \|f\|_{L^1} \cdot \varepsilon(\Phi, u)^{1/p}\right\}. \tag{66}$$

Theorem 3.1 follows easily from averaging each side over u.

Proof of Lemma 3.3. Without loss of generality, suppose $u = e_1 = (1, 0, \dots, 0)$; then

$$\varepsilon(\Psi, u) = \max_{(x,y)\in\mathbb{R}\times\mathbb{R}^{d-1}} |x - \psi_1(x, y)|. \tag{67}$$

For brevity, if $h : \mathbb{R}^d \to \mathbb{R}$ is a function of d variables, let $\mathcal{P}h = \mathcal{P}_{e_1} h$. That is,

$$(\mathcal{P}h)(x) = \int_{\mathbb{R}^{d-1}} h(x, y) dy. \tag{68}$$

For $(x, y) \in \mathbb{R}\times\mathbb{R}^{d-1}$, let $I_{(x,y)}$ be the interval $[x, \psi_1(x, y)]$ when $x \le \psi_1(x, y)$, and $[\psi_1(x, y), x]$ when $x > \psi_1(x, y)$; and let $\chi(x, y, t)$ be 1 if $t \in I_{(x,y)}$, and 0 otherwise; that is, $\chi(x, y, t) = 1$ if either $x \le t \le \psi_1(x, y)$ or $\psi_1(x, y) \le t \le x$.

Step 1. We will show that for all $(x, y) \in \mathbb{R} \times \mathbb{R}^{d-1}$,

$$\int \chi(x, y, t)\, dt \le \varepsilon(\Psi, u), \tag{69}$$

and for all t,

$$\int \sup_y \chi(x, y, t)\, dx \le 2\varepsilon(\Psi, u). \tag{70}$$

For the first inequality, for fixed (x, y), suppose without loss of generality that $x \le \psi_1(x, y)$. Then $\chi(x, y, t) = 1$ if and only if $x \le t \le \psi_1(x, y)$, and so

$$\int \chi(x, y, t)\, dt = |\psi_1(x, y) - x| \le \varepsilon(\Psi, u), \tag{71}$$

which is (69).

For the second inequality: for any x and t, $\sup_y \chi(x, y, t) = 1$ if and only if there exists a vector y such that either $x \le t \le \psi_1(x, y)$ or $\psi_1(x, y) \le t \le x$. In this case, since $|x - \psi_1(x, y)| \le \varepsilon(\Psi, u)$, we must also have $|x - t| \le \varepsilon(\Psi, u)$, and so x lies in the interval $[t - \varepsilon(\Psi, u), t + \varepsilon(\Psi, u)]$ of length $2\varepsilon(\Psi, u)$; hence

$$\int \sup_y \chi(x, y, t)\, dx \le 2\varepsilon(\Psi, u), \tag{72}$$

which is (70).

Step 2. We will prove that

$$\mathrm{C}_p(\mathcal{P}_u f, \mathcal{P}_u f_\Phi) \le 2^{(p-1)/p} \cdot \|\mathcal{P}_u(|f|)\|_{L^p} \cdot \varepsilon(\Phi, u). \tag{73}$$

Let $G \in \mathcal{A}_0$, with derivative $g = G'$ satisfying $\|g\|_{L^q} \le 1$. Performing the change of variables $w = \Phi(x, y)$ gives

$$\begin{aligned}
\int_{\mathbb{R}} G(x)(\mathcal{P}f_\Phi)(x)\, dx &= \int_{\mathbb{R}} G(x) \int_{y:(x,y)\in B} f(\Phi(x, y))|\det(\nabla\Phi(x, y))|\, dy\, dx \\
&= \int_B G(x) f(\Phi(x, y))|\det(\nabla\Phi(x, y))|\, dy\, dx \\
&= \int_A G(\psi_1(w)) f(w)\, dw \\
&= \int_{\mathbb{R}} \int_{y:(x,y)\in A} G(\psi_1(x, y)) f(x, y)\, dy\, dx
\end{aligned} \tag{74}$$

and similarly,

$$\int_{\mathbb{R}} G(x)(\mathcal{P}f)(x)\,dx = \int_{\mathbb{R}} \int_{y:(x,y)\in A} G(x)f(x,y)\,dy\,dx. \tag{75}$$

Combining (74) and (75), and applying Hölder's inequality,

$$\begin{aligned}
\int_{\mathbb{R}} G(x)((\mathcal{P}f)(x) - (\mathcal{P}f_\Phi)(x))\,dx &= \int\int [G(x) - G(\psi_1(x,y))]f(x,y)\,dy\,dx \\
&\le \int \left(\sup_y |G(x) - G(\psi_1(x,y))|\right)\left(\int_{\mathbb{R}^{d-1}} |f(x,y)|\,dy\right)dx \\
&= \int \left(\sup_y |G(x) - G(\psi_1(x,y))|\right)\mathcal{P}(|f|)(x)dx \\
&\le \|\mathcal{P}(|f|)\|_{L^p} \cdot \left\|\sup_y |G(x) - G(\psi_1(x,y))|\right\|_{L^q(dx)}, \qquad (76)
\end{aligned}$$

where $1/p + 1/q = 1$.

We will show that

$$\left\|\sup_y |G(x) - G(\psi_1(x,y))|\right\|_{L^q(dx)} \le 2^{(p-1)/p}\,\varepsilon(\Psi,u), \tag{77}$$

which yields (73) by applying Proposition 2.6. We have

$$|G(x) - G(\psi_1(x,y))| = \left|\int_x^{\psi_1(x,y)} g(t)dt\right| = \left|\int g(t)\chi(x,y,t)dt\right|. \tag{78}$$

Suppose temporarily that $1 < p, q < \infty$. Applying Hölder's inequality and using (69) and (70), we get

$$\begin{aligned}
\int \left(\sup_y |G(x) - G(\psi_1(x,y))|\right)^q dx &= \int \sup_y |G(x) - G(\psi_1(x,y))|^q dx \\
&= \int \sup_y \left|\int g(t)\chi(x,y,t)dt\right|^q dx \\
&\le \int \sup_y \left(\int |g(t)|^q \chi(x,y,t)dt\right)\left(\int \chi(x,y,t)\,dt\right)^{q/p} dx \\
&\le \varepsilon(\Psi,u)^{q/p} \int \sup_y \int |g(t)|^q \chi(x,y,t)\,dt\,dx \\
&\le \varepsilon(\Psi,u)^{q/p} \int\int |g(t)|^q \sup_y \chi(x,y,t)\,dt\,dx \\
&= \varepsilon(\Psi,u)^{q/p} \int |g(t)|^q \int \sup_y \chi(x,y,t)\,dx\,dt \\
&\le 2\varepsilon(\Psi,u)^{q/p+1} \int |g(t)|^q\,dt, \qquad (79)
\end{aligned}$$

and so taking the q-th root we get the bound

$$\left\|\sup_y |G(x) - G(\psi_1(x,y))|\right\|_{L^q(dx)} \le 2^{1/q}\varepsilon(\Psi,u)^{1/p+1/q}\|g\|_{L^q} \le 2^{(p-1)/p}\varepsilon(\Psi,u). \tag{80}$$

Now suppose $p = 1$ and $q = \infty$. Then from (69),

$$\begin{aligned}
\left\| \sup_y |G(x) - G(\psi_1(x,y))| \right\|_{L^\infty(dx)} &= \sup_{x,y} |G(x) - G(\psi_1(x,y))| \\
&= \sup_{x,y} \left| \int g(t)\chi(x,y,t)dt \right| \\
&\leq \|g\|_{L^\infty} \sup_{x,y} \int \chi(x,y,t)\, dt \\
&\leq \|g\|_{L^\infty} \varepsilon(\Psi, u) \\
&\leq \varepsilon(\Psi, u). \qquad (81)
\end{aligned}$$

Finally, suppose $p = \infty$ and $q = 1$. Then from (70),

$$\begin{aligned}
\left\| \sup_y |G(x) - G(\psi_1(x,y))| \right\|_{L^1(dx)} &= \int \sup_y |G(x) - G(\psi_1(x,y))|\, dx \\
&= \int \sup_y \left| \int g(t)\chi(x,y,t)dt \right| dx \\
&\leq \int \int |g(t)| \sup_y \chi(x,y,t)\, dt\, dx \\
&= \int |g(t)| \int \sup_y \chi(x,y,t)\, dx\, dt \\
&\leq \|g\|_{L^1} \sup_t \int \sup_y \chi(x,y,t)\, dx \\
&\leq \|g\|_{L^1} 2\varepsilon(\Psi, u) \\
&= 2\varepsilon(\Psi, u). \qquad (82)
\end{aligned}$$

This completes the proof of (77), and hence proves (73).

Step 3. To conclude the proof, we will prove the inequality

$$\mathrm{C}_p(\mathcal{P}_u f, \mathcal{P}_u f_\Phi) \leq \|f\|_{L^1} \cdot \varepsilon(\Psi, u)^{1/p}. \qquad (83)$$

Let $G \in \mathcal{A}_0$, with derivative $g = G'$ satisfying $\|g\|_{L^q} \leq 1$. From (76), taking $p = 1$ and $q = \infty$,

$$\int_{\mathbb{R}} G(x)((\mathcal{P}f)(x) - (\mathcal{P}f_\Phi)(x))\, dx \leq \|f\|_{L^1} \sup_{x,y} |G(x) - G(\psi_1(x,y))|, \qquad (84)$$

and so by using Proposition 2.6, it is enough to show that for all (x, y),

$$|G(x) - G(\psi_1(x,y))| \leq \varepsilon(\Psi, u)^{1/p}. \qquad (85)$$

Hölder's inequality yields

$$\begin{aligned}
|G(x) - G(\psi_1(x,y))| &= \left| \int g(t)\chi(x,y,t)dt \right| \\
&\leq \|g\|_{L^q} \left(\int \chi(x,y,t)^p\, dt \right)^{1/p} \\
&\leq \left(\int \chi(x,y,t)\, dt \right)^{1/p} \\
&\leq \varepsilon(\Psi, u)^{1/p}, \qquad (86)
\end{aligned}$$

where the last inequality follows from (69). This completes the proof.

□

Next, we will consider special cases for which quantitatively tighter bounds can be shown.

3.2 Sharper bounds in special cases

In this section, we consider specific classes of deformations, and prove sharper bounds than (49) and (50) from Theorem 3.1.

3.2.1 Rotations in 2D

We consider the case where $A = B = \mathbb{D} \subset \mathbb{R}^2$, the open unit disc centered at $(0, 0)$. If Φ is a rotation around the origin by angle θ, then the corresponding maximum displacement is $\varepsilon(\Phi) = 2\sin(\theta/2)$, and so the bound (49) in Theorem 3.1 is

$$\mathrm{SC}_{\eta,p}(f, f_\Phi) \leq 2^{(p-1)/p} \cdot \|f\|_{M_\eta^p} \cdot 2\sin(\theta/2). \tag{87}$$

(We do not consider the bound (50), as for this choice of Φ it is never stronger than (49).)

We can prove a sharper estimate:

Theorem 3.4. *Let $1 \leq p \leq \infty$, and $f : \mathbb{R}^2 \to \mathbb{R}$ be in $L^p(\mathbb{D})$. Suppose $0 \leq \theta < \pi$, and define f_θ by*

$$f_\theta(x, y) = f(x\cos(\theta) + y\sin(\theta), y\cos(\theta) - x\sin(\theta)). \tag{88}$$

Then for any probability distribution η over $\mathbb{S}^{d-1}$,

$$\mathrm{SC}_{\eta,p}(f, f_\theta) \leq \|f\|_{M_\eta^p} \cdot \Delta_p(\theta), \tag{89}$$

where

$$\Delta_p(\theta) = \begin{cases} 2\sin(\theta/2) \cdot (2\cos(\theta/2))^{(p-1)/p}, & \text{if } 0 \leq \theta < \pi/2 \\ 2\sin(\theta/2)^{1/p}, & \text{if } \pi/2 \leq \theta < \pi \end{cases}. \tag{90}$$

The result follows from the following lemma:

Lemma 3.5. *Using the notation from the statement of Theorem 3.4, if u is any unit vector in $\mathbb{R}^2$, then*

$$\mathrm{C}_p(\mathcal{P}_u f, \mathcal{P}_u f_\theta) \leq \|\mathcal{P}_u(|f|)\|_{L^p} \cdot \Delta_p(\theta). \tag{91}$$

Theorem 3.4 follows immediately by taking the p-th power and averaging over all u.

Proof of Lemma 3.5. Without loss of generality, suppose $u = (1, 0)$. An identical proof to that of Lemma 3.3 may be applied by replacing the bound $\sup_t \int \sup_y \chi(x, y, t)dx \leq 2\varepsilon(\Phi)$ from (70) with the bound

$$\int \sup_y \chi(x, y, t)dx \leq \begin{cases} 2\sin(\theta), & \text{if } 0 \leq \theta < \pi/2 \\ 2, & \text{if } \pi/2 \leq \theta < \pi \end{cases} \tag{92}$$

for all $|t| < 1$. Indeed, when $0 \leq \theta < \pi/2$, we can then replace (77) with the upper bound

$$\begin{aligned} \left\| \sup_y |G(x) - G(\psi_1(x, y))| \right\|_{L^q(dx)} &\leq (2\sin(\theta/2))^{1/p} \cdot (2\sin(\theta))^{1/q} \\ &= (2\sin(\theta/2))^{1/p} \cdot (4\sin(\theta/2)\cos(\theta/2))^{1/q} \\ &= 2\sin(\theta/2) \cdot (2\cos(\theta/2))^{1/q} \\ &= 2\sin(\theta/2) \cdot (2\cos(\theta/2))^{(p-1)/p} \end{aligned} \tag{93}$$

whereas when $\pi/2 \leq \theta < \pi$ the bound becomes

$$\left\| \sup_y |G(x) - G(\psi_1(x, y))| \right\|_{L^q(dx)} \leq (2\sin(\theta/2))^{1/p} \cdot 2^{1/q} = 2\sin(\theta/2)^{1/p}. \tag{94}$$

The bound $\int \sup_y \chi(x, y, t)dx \leq 2$ is immediate, since the integrand is bounded by 1 and the integral is over $|x| \leq 1$. Hence it remains to show that

$$\int \sup_y \chi(x, y, t)dx \leq 2\sin(\theta) \tag{95}$$

whenever $0 \leq \theta < \pi/2$.

Let $c = \cos(\theta)$ and $s = \sin(\theta)$. Note that in this case, $c \geq 0$ and $s \geq 0$; and the rotation $\Phi(x, y) = (cx+sy, cy-sx)$, with inverse $\Psi(x, y) = (cx - sy, cy + sx)$.

Take $|t| < 1$, and suppose, without loss of generality, that $0 \leq t \leq 1$. It is enough to show

$$\int \sup_y \chi(x, y, t)\, dx \leq \begin{cases} 2s\sqrt{1-t^2}, & \text{if } 0 \leq t < c \\ 1 - ct + s\sqrt{1-t^2}, & \text{if } c \leq t \leq 1 \end{cases}. \tag{96}$$

Indeed, the right side of (96) is decreasing in t and so is maximized when $t = 0$, which yields the desired bound (95).

We now show (96). For $0 \leq t \leq 1$, let S_t denote the set of all x, $|x| \leq 1$, satisfying $\sup_y \chi(x, y, t) = 1$. Then $\int \sup_y \chi(x, y, t)\, dx = |S_t|$.

Lemma 3.6. *Let* $0 \leq t \leq 1$.

1. *Suppose* $t \leq x$. *Then* $x \in S_t$ *if and only if* $cx - s\sqrt{1-x^2} \leq t$.
2. *Suppose* $x \leq t$. *Then* $x \in S_t$ *if and only if* $t \leq cx + s\sqrt{1-x^2}$.

Proof. First suppose that $t \leq x$. If $x \in S_t$, then there exists y with $\chi(x, y, t) = 1$, that is, for which $(x, y) \in \mathbb{D}$ and $\psi_1(x, y) = cx - sy \leq t \leq x$. Since $cx - sy$ only gets smaller as y grows, we can always take $y = \sqrt{1-x^2}$. The converse is immediate.

Next suppose that $x \leq t$. If $x \in S_t$, then there exists y for which $(x, y) \in \mathbb{D}$ and $x \leq t \leq cx - sy$. Since $cx - sy$ only gets bigger as y shrinks, we can always take $y = -\sqrt{1-x^2}$. Again, the converse is immediate.

□

Lemma 3.7. *Let* $0 \leq t \leq 1$. *Then for all* $x \in S_t$, $x \geq ct - s\sqrt{1-t^2}$.

Proof. We will break this into two cases, depending on whether $x \leq t$ or $x > t$.

Case 1. $x \leq t$. By Lemma 3.6, $cx + s\sqrt{1-x^2} \geq t$. Therefore, $s\sqrt{1-x^2} \geq t - cx$ and since we assume $x \leq t$, $t - cx \geq 0$; therefore, squaring each side gives

$$s^2 - s^2x^2 \geq t^2 + c^2x^2 - 2ctx, \tag{97}$$

and hence

$$x^2 - 2ctx + t^2 - s^2 \leq 0, \tag{98}$$

and since the roots of the polynomial are $x = ct \pm s\sqrt{1-t^2}$, in particular it follows that $x \geq ct - s\sqrt{1-t^2}$.

Case 2. $x > t$. In this case, the result follows immediately from the inequality $ct - s\sqrt{1-t^2} \leq ct \leq t < x$.

□

Lemma 3.8. *Suppose that* $0 \leq t \leq c$. *Then all* $x \in S_t$ *satisfy* $x \leq ct + s\sqrt{1-t^2}$.

Proof. First, note that the assumption $0 \leq t \leq c$ implies that $t \leq ct + s\sqrt{1-t^2}$. Indeed, the latter inequality is equivalent to $t(1-c) \leq s\sqrt{1-t^2}$, and which is equivalent to $t^2 + t^2c^2 - 2ct^2 \leq s^2 - t^2s^2$, which is equivalent to $2t^2(1-c) \leq s^2 = 1 - c^2 = (1-c)(1+c)$, which is equivalent to $2t^2 \leq 1 + c$, which is immediate from $0 \leq t \leq c \leq 1$.

Now, let $x \in S_t$. If $x \leq t$, then $x \leq ct + s\sqrt{1-t^2}$ as well, by what we have just shown. Therefore, we can suppose instead that $x > t$. By Lemma 3.6, $cx - s\sqrt{1-x^2} \leq t$. Note that $cx - s\sqrt{1-x^2}$ is an increasing function of x. Hence, if $x > t$ satisfies

$$cx - s\sqrt{1-x^2} = t, \tag{99}$$

then for all $x' > x$, we have $x' > t$ and $cx' - s\sqrt{1-(x')^2} > t$, and hence by Lemma 3.6 $x' \notin S_t$. For $x > t$ satisfying (99), we have

$$\begin{aligned} & cx - t = s\sqrt{1-x^2} \\ \implies\ & c^2x^2 + t^2 - 2ctx = s^2 - s^2x^2 \\ \implies\ & x^2 - 2ctx + t^2 - s^2 = 0 \\ \implies\ & x = ct \pm s\sqrt{1-t^2} \\ \implies\ & x = ct + s\sqrt{1-t^2}, \end{aligned} \tag{100}$$

where the last implication is because $x > t$. This completes the proof.

□

From Lemmas 3.7 and 3.8, when $0 \le t < c$, all $x \in S_t$ lie between $ct - s\sqrt{1-t^2}$ and $ct + s\sqrt{1-t^2}$; hence $|S_t| \le 2s\sqrt{1-t^2}$. On the other hand, when $t \ge c$, by Lemma 3.7 all $x \in S_t$ must lie between $ct - s\sqrt{1-t^2}$ and 1; hence $|S_t| \le 1 - ct + s\sqrt{1-t^2}$. This concludes the proof of (96), and hence of Lemma 3.5.

□

3.2.2 Monotonically increasing deformations in 1D

When $d = 1$, the bounds (49) and (50) in Theorem 3.1 become

$$\mathrm{C}_p(f, f_\Phi) \le 2^{(p-1)/p} \cdot \|f\|_{L^p} \cdot \varepsilon(\Phi). \tag{101}$$

Now we show that the factor of $2^{(p-1)/p}$ can be removed when the deformation Φ is monotonically increasing.

Theorem 3.9. *Let $1 \le p \le \infty$. Let I and J be non-empty, bounded, open intervals in $\mathbb{R}$, f be in $L^p(I)$, $\Phi : J \to I$ be a C^1 bijection with inverse Ψ and $\Phi'(x) > 0$ on J, and $f_\Phi(x) = f(\Phi(x))\Phi'(x))$ on J, and 0 elsewhere. Then*

$$\mathrm{C}_p(f, f_\Phi) \le \|f\|_{L^p} \cdot \varepsilon(\Phi). \tag{102}$$

Remark 13. To see that the monotonocity of Φ is required for this sharper bound, consider the following example. Fix $\eta > \delta > 0$, and let f be defined by

$$f(x) = \begin{cases} 1, & \text{if } -\eta \le x < 0, \\ -1, & \text{if } 0 \le x \le \eta, \\ 0, & \text{otherwise.} \end{cases} \tag{103}$$

Let $\Phi : [-\delta, \delta] \to [-\eta, \eta]$ be defined by $\Phi(x) = -(\eta/\delta)x$. Then

$$f_\Phi(x) = \begin{cases} -\eta/\delta, & \text{if } -\delta \le x < 0, \\ \eta/\delta, & \text{if } 0 \le x \le \delta, \\ 0, & \text{otherwise.} \end{cases} \tag{104}$$

Then it is straightforward to verify that $\varepsilon(\Phi) = \eta + \delta$, $\|f\|_{L^\infty} = 1$, and $C_\infty(f, f_\Phi) = 2\eta$. By taking $\delta \to 0$, we see that the bound $C_\infty(f, f_\Phi) \le 2\|f\|_{L^\infty}\varepsilon(\Phi)$ is tight.

Proof of Theorem 3.9. Let I_x be the interval $[x, \Psi(x)]$ if $x \le \Psi(x)$, and $[\Psi(x), x]$ if $\Psi(x) \le x$. Let $\chi(x, t)$ be 1 if $t \in I_x$, and 0 otherwise. Then an identical proof to that of Lemma 3.3 may be applied if we show that

$$\sup_t \int \chi(x, t) dx \le \varepsilon(\Phi), \tag{105}$$

in place of the bound (70).

Take any $t \in I$, and suppose that there is some $x \le t$ with $t \in I_x$; note that for such x, $I_x = [x, \Psi(x)]$, and so $x \le \Psi(x)$. Let x^* be the smallest such x. Then $x^* \le t \le \Psi(x^*)$. We claim that for all $x > t$, $t \notin I_x$. Indeed, since Ψ is increasing and $x > t \ge x^*$, we have $\Psi(x) > \Psi(x^*) \ge t$. Since both $x > t$ and $\Psi(x) > t$, t does not lie in I_x, as claimed.

Consequently, all x for which t lies in I_x are contained inside the interval $[x^*, t]$. Since $x^* \le t \le \Psi(x^*)$ and $|x^* - \Psi(x^*)| \le \varepsilon(\Phi)$, it follows that $|t - x^*| \le \varepsilon(\Phi)$ too. Furthermore, if $x > t$, then $\chi(x, t) = 0$ since $t \notin I_x$; and since x^* is the smallest x for which $t \in I_x$, if $x < x^*$ then $t \notin I_x$, hence $\chi(x, t) = 0$. Therefore,

$$\int \chi(x, t) dx \le \int_{x^*}^{t} 1 dx = |t - x^*| \le \varepsilon(\Phi). \tag{106}$$

Analogous reasoning yields the same bound in the case that there exists $x \ge t$ with $t \in I_x$. This completes the proof.

□

3.2.3 Translations

We consider the case where $\Phi(x) = x + v$, where v is a fixed vector. For simplicity, we only describe the case where η is the uniform measure, though the results easily generalize. In this case, the bounds (49) and (50) from Theorem 3.1 are, respectively,

$$\mathrm{SC}_p(f, f_\Phi) \leq 2^{(p-1)/p} \cdot \|f\|_{M^p} \cdot |v| \tag{107}$$

and

$$\mathrm{SC}_p(f, f_\Phi) \leq 2^{(p-1)/p} \cdot K_p \cdot \|f\|_{M^{p,\infty}} \cdot |v|, \tag{108}$$

where

$$K_p = \left(\int_{\mathbb{S}^{d-1}} |u_1|^p \, du \right)^{1/p}. \tag{109}$$

We show that the factor $2^{(p-1)/p}$ can be removed:

Theorem 3.10. *Let $1 \leq p \leq \infty$. Let A be a non-empty, bounded, open set in $\mathbb{R}^d$, and f be in $L^p(A)$. For a fixed vector v, let $f_v(x) = f(x + v)$. Then*

$$\mathrm{SC}_p(f, f_v) \leq \|f\|_{M^p} \cdot |v| \tag{110}$$

and

$$\mathrm{SC}_p(f, f_v) \leq K_p \cdot \|f\|_{M^{p,\infty}} \cdot |v|, \tag{111}$$

where K_p is defined in (109).

Remark 14. When $p = 1$, $\mathrm{SC}_1(f, f_v) = \mathrm{SW}_1(f, f_v)$, and so the bound (111) when $p = 1$ and $d = 2$ matches the $p = 1$ case of Theorem 2 of [65].

Proof of Theorem 3.10. By a straightforward calculation, for any $u \in \mathbb{S}^{d-1}$, $(\mathcal{P}_u f_v)(t) = (\mathcal{P}_u f)(t + \langle v, u \rangle)$. Consequently, by Theorem 3.9,

$$\mathrm{C}_p(\mathcal{P}_u f, \mathcal{P}_u f_v) \leq \|\mathcal{P}_u f\|_{L^p} \cdot |\langle v, u \rangle|. \tag{112}$$

Then

$$\mathrm{SC}_p(f, f_v)^p \leq \int_{\mathbb{S}^{d-1}} \|\mathcal{P}_u f\|_{L^p}^p \cdot |\langle v, u \rangle|^p \, du \leq |v|^p \cdot \int_{\mathbb{S}^{d-1}} \|\mathcal{P}_u f\|_{L^p}^p \, du \leq \|f\|_{M^p} \cdot |v|^p, \tag{113}$$

which is (110); and

$$\mathrm{SC}_p(f, f_v)^p \leq \sup_{u \in \mathbb{S}^{d-1}} \|\mathcal{P}_u f\|_{L^p}^p \cdot \int_{\mathbb{S}^{d-1}} |\langle v, u \rangle|^p \, du \leq K_p \cdot \|f\|_{M^{p,\infty}}^p \cdot |v|^p, \tag{114}$$

proving (111).

$\square$

3.2.4 Dilations

Suppose $\mathbb{B} \subset \mathbb{R}^d$ is the open unit ball in $\mathbb{R}^d$ centered at 0. Let $\Phi(w) = \alpha w$, where $\alpha > 1$. Then $\varepsilon(\Phi) = \alpha - 1$, and so bound (49) Theorem 3.1 is

$$\mathrm{SC}_{\eta,p}(f, f_\Phi) \leq 2^{(p-1)/p} \cdot \|f\|_{M_\eta^p} \cdot (\alpha - 1). \tag{115}$$

(We do not consider the bound (50), as for this choice of Φ it is never stronger than (49).)

We can prove a sharper estimate:

Theorem 3.11. *Let $1 \le p \le \infty$, and let f be in $L^p(\mathbb{B})$. Suppose $\alpha > 1$, and define f_α by $f_\alpha(w) = \alpha f(\alpha w)$. Then for any probability distribution η over $\mathbb{S}^{d-1}$,*

$$\mathrm{SC}_{\eta,p}(f, f_\alpha) \le \|f\|_{M^p_\eta} \cdot \frac{\alpha - 1}{\alpha^{(p-1)/p}}. \tag{116}$$

The result follows from the following lemma:

Lemma 3.12. *Using the notation from the statement of Theorem 3.11,*

$$\mathrm{C}_p(\mathcal{P}_u f, \mathcal{P}_u f_\alpha) \le \|\mathcal{P}_u(|f|)\|_{L^p} \cdot \frac{\alpha - 1}{\alpha^{(p-1)/p}}. \tag{117}$$

Theorem 3.4 follows immediately by taking the p-th power and averaging over all u.

Proof of Lemma 3.12. Without loss of generality, suppose $u = (1, 0, \ldots, 0)$. It is enough to show that for all $|t| < 1$,

$$\int \sup_{y \in \mathbb{R}^{d-1}} \chi(x, y, t)\, dx \le \frac{\alpha - 1}{\alpha}; \tag{118}$$

this estimate can then be used in place of (70) in the proof of Lemma 3.3.

Without loss of generality, suppose $0 \le t < 1$. Let S_t denote the set of all x, $|x| < 1$, satisfying $\sup_{y \in \mathbb{R}^{d-1}} \chi(x, y, t) = 1$. If $x \in S_t$, then $x < t < \alpha x$; since $t \ge 0$, this restricts $x \ge 0$ as well, and $S_t = (t/\alpha, t)$, so $|S_t| = t(1 - 1/\alpha)$, which is maximized at $t = 1$; thus

$$\int \sup_{y \in \mathbb{R}^{d-1}} \chi(x, y, t)\, dx \le 1 - 1/\alpha = \frac{\alpha - 1}{\alpha}, \tag{119}$$

as claimed. Using this estimate in place of the bound $\int \sup_{y \in \mathbb{R}^{d-1}} \chi(x, y, t)\, dx \le 2\varepsilon(\Phi)$ gives the final bound

$$\mathrm{C}(\mathcal{P}_u f, \mathcal{P}_u f_\Phi) \le \|\mathcal{P}_u(|f|)\|_{L^p} (\alpha - 1)^{1/p} \left(\frac{\alpha - 1}{\alpha}\right)^{1/q} = \|\mathcal{P}_u(|f|)\|_{L^p} \cdot \frac{(\alpha - 1)}{\alpha^{(p-1)/p}}, \tag{120}$$

completing the proof.

□

3.3 Convolutions

In this section, we remark on the behavior of the sliced Cramér distance after convolution of its input functions. This situation occurs commonly in signal and image processing, where one typically observes signals that have been convolved with a function induced from the measurement apparatus. The bound we state here is analogous to Theorem 2.4 and Corollary 2.5 for Wasserstein and sliced Wasserstein distances, respectively. Our result is a slight generalization of the first part of Theorem 5 in the prior work [80], where it is stated in terms of random variables (and hence only applies to non-negative functions) and specialized to the case $p = 2$; we think it is valuable to explicitly state and prove the result for general p and reformulate it in terms of convolution, where its applicability to signal and image processing may be more evident; furthermore, our bound is sharper when the convoluton kernel takes on negative values, as can occur in a range of scientific imaging applications.

Theorem 3.13. *Let $1 \le p \le \infty$. Suppose f and g are in $L^p(\mathbb{R}^d)$ and compactly supported, w is in $L^1(\mathbb{R}^d)$ and compactly supported, and η is a probability measure over $\mathbb{S}^{d-1}$. Then*

$$\mathrm{SC}_{\eta,p}(f * w, g * w) \le \left(\int_{\mathbb{S}^{d-1}} \|\mathcal{P}_u w\|_{L^1} d\eta(u)\right) \cdot \mathrm{SC}_{\eta,p}(f, g). \tag{121}$$

Remark 15. When $w \ge 0$, the conclusion of Theorem 3.13 becomes

$$\mathrm{SC}_{\eta,p}(f * w, g * w) \le \|w\|_{L^1} \cdot \mathrm{SC}_{\eta,p}(f, g), \tag{122}$$

which is essentially what appears (for $p = 2$) in [80]. However, in certain scientific imaging applications, such as cryo-electron microscopy, where w is the point-spread function of the imaging apparatus, w may take negative values, in which case the bound (121) is sharper than (122).

Proof. We first show the result for $d = 1$, that is, for when f, g and w are functions on $\mathbb{R}$. That is, we will show that

$$\mathrm{C}_p(f * w, g * w) \leq \|w\|_{L^1} \cdot \mathrm{C}_p(f, g). \tag{123}$$

(Note that this case essentially appears in Theorem 2 of [6]; we provide an alternate proof for the reader's convenience.) Let H be in $\mathcal{A}_0$, with $\|H'\|_{L^q} = 1$. For any s, $\|H'(u + s)\|_{L^q(du)} = 1$ too. Then using Proposition 2.6,

$$\begin{aligned}
\langle f * w - g * w, H \rangle &= \int ((f - g) * w)(t) H(t)\, dt \\
&= \int \int (f - g)(t - s) w(s)\, ds\, H(t)\, dt \\
&= \int w(s) \int (f - g)(t - s) H(t)\, dt\, ds \\
&= \int w(s) \int (f - g)(u) H(u + s)\, du\, ds \\
&\leq \|w\|_{L^1} \sup_s \left| \int (f - g)(u) H(u + s)\, du \right| \\
&\leq \|w\|_{L^1} \mathrm{C}_p(f, g),
\end{aligned} \tag{124}$$

and (123) now follows by taking the supremum over all such H and invoking Proposition 2.6.

We now turn to general $d \geq 1$. First, observe that $\mathcal{P}_u(w * h) = (\mathcal{P}_u w) * (\mathcal{P}_u h)$: indeed, by the Fourier slice theorem,

$$\begin{aligned}
(\mathcal{P}_u(w * h))\widehat{\ }(\xi) &= \widehat{(w * h)}(\xi u) \\
&= \widehat{w}(\xi u) \widehat{h}(\xi u) \\
&= \widehat{(\mathcal{P}_u w)}(\xi) \widehat{(\mathcal{P}_u h)}(\xi) \\
&= [(\mathcal{P}_u w) * (\mathcal{P}_u h)]\widehat{\ }(\xi),
\end{aligned} \tag{125}$$

and so, taking inverse Fourier transforms, $\mathcal{P}_u(w * h) = (\mathcal{P}_u w) * (\mathcal{P}_u h)$.

From the 1D bound, we then have

$$\begin{aligned}
\mathrm{C}_p(\mathcal{P}_u(w * f), \mathcal{P}_u(w * g)) &= \mathrm{C}_p((\mathcal{P}_u w) * (\mathcal{P}_u f), (\mathcal{P}_u w) * (\mathcal{P}_u g)) \\
&\leq \|\mathcal{P}_u w\|_{L^1} \mathrm{C}_p(\mathcal{P}_u f, \mathcal{P}_u g)
\end{aligned} \tag{126}$$

Taking p-th powers and integrating over u gives the result.

□

4 Discretizations and robustness to noise

In this section, we will describe Fourier-based discretizations of the Cramér distance and the 2D sliced Cramér distance with respect to the uniform measure over $\mathbb{S}^1$, between functions with equal integrals, and analyze their robustness to additive, heteroscedastic sub-Gaussian noise. More precisely, we will show that, given vectors of noisy samples from two smooth functions, the discrete distance approximates the distances between the smooth functions only, removing the effect of the noise as the number of samples grows. These results are to be expected; indeed, the Cramér distance itself involves applying a smoothing filter to each input, which, by averaging the samples, naturally has a denoising effect. The denoising property is also of interest in contrasting with Wasserstein and sliced Wasserstein distances, which, because they are defined between probability measures, do not naturally induce distances between vectors sampled from a signal-plus-noise model without modification of the definition.

4.1 Robustness to noise in 1D

We define a discrete approximation to the 1D Volterra norm, which then yields an approximation to the Cramér distance. Let $a < b$ and let $L = b - a$ be the interval length. Let n a positive integer; we will assume for simplicity that n is even. Let $x \in \mathbb{R}^n$; the reader should think of the entries $x[j]$ of x as (possibly noisy) samples from a function f on $[a,b]$, that is, $x[j] \approx f(t_j)$, $j = 0, \dots, n-1$, where $t_j = a + jL/n$.

Define the values $\alpha[k]$ (the normalized discrete Fourier coefficients of x) by

$$\alpha[k] = \frac{L}{n} \sum_{j=0}^{n-1} x[j] e^{-2\pi \mathrm{i} k t_j / L}, \tag{127}$$

for integers k. Then for $-n/2+1 \le k \le n/2-1$, $\widehat{f}(k/L) \approx \alpha[k]$.

When $0 < |k| < n/2$, define

$$\beta[k] = \frac{\alpha[k]}{2\pi \mathrm{i} k/L}, \tag{128}$$

and when $k = 0$, define

$$\beta[0] = -\sum_{0<|k|<n/2} \beta[k] e^{2\pi \mathrm{i} k a/L}. \tag{129}$$

Then the $\beta[k]$ approximate the Fourier coefficients of $\mathcal{V}f$: $\widehat{(\mathcal{V}f)}(k/L) \approx \beta[k]$.

Define the function $\nu_x(t)$ by

$$\nu_x(t) = \frac{1}{L} \sum_{k=-n/2+1}^{n/2-1} \beta[k] e^{2\pi \mathrm{i} t k/L}. \tag{130}$$

(Note that we do not define $\beta[\pm n/2]$, because the terms they would contribute to $\nu_x(t)$ would either be purely imaginary or 0, depending on the convention.) Then for all t, $\nu_x(t) \approx (\mathcal{V}f)(t)$.

We then define the discrete Volterra p-norm of the vector x as follows:

$$\mathrm{V}_p(x) = \left(\frac{L}{n} \sum_{j=0}^{n-1} |\nu_x(t_j)|^p \right)^{1/p} \tag{131}$$

when $1 \le p < \infty$, and

$$\mathrm{V}_\infty(x) = \max_{0 \le j \le n-1} |\nu_x(t_j)| \tag{132}$$

when $p = \infty$. Given two vectors x and y in $\mathbb{R}^n$, we then define their discrete Cramér distance as

$$\widehat{\mathrm{C}}_p(x,y) = \mathrm{V}_p(x-y). \tag{133}$$

Remark 16. Using the Fast Fourier Transform (FFT) and the inverse FFT (IFFT) to evaluate the $\alpha[k]$ and $\nu_x(t_j)$, respectively, the entire computation described here can be performed at cost $O(n \log n)$.

We can now state the main result from this section, which says that the discrete Cramér distance is robust to additive heteroscedastic sub-Gaussian noise:

Theorem 4.1. *Suppose f and g are C^r functions on $\mathbb{R}$, $r > 1$, that are supported on $[a,b]$, and satisfy $\int_a^b f = \int_a^b g$. Let $Z[0], Z[1], \dots, Z[n-1]$, $\widetilde{Z}[0], \dots, \widetilde{Z}[n-1]$ be independent, mean-zero sub-Gaussian random variables with sub-Gaussian norms $\sigma_j = \|Z[j]\|_{\psi_2}$ and $\widetilde{\sigma}_j = \|\widetilde{Z}[j]\|_{\psi_2}$; and suppose $\sigma > 0$ satisfies*

$$\frac{1}{n} \sum_{j=0}^{n-1} \sigma_j^2 + \frac{1}{n} \sum_{j=0}^{n-1} \widetilde{\sigma}_j^2 \le \sigma^2, \tag{134}$$

for all n. Let X_n and Y_n be vectors in $\mathbb{R}^n$ with entries $X_n[j] = f(t_j) + Z[j]$ and $Y_n[j] = g(t_j) + \widetilde{Z}[j]$. Then:

1. *Expected error. For $1 \le p < \infty$, there are values A and $B = B(p,f)$ such that for all $\sigma \ge 0$ and $n \ge 2$,*

$$\mathbb{E}\big|\widehat{\mathrm{C}}_p(X_n,Y_n) - \mathrm{C}_p(f,g)\big| \le AL^{1+1/p}\sqrt{p}\frac{\sigma}{\sqrt{n}} + B\max\left\{\frac{\log(n)}{n^r},\frac{1}{n^2}\right\}. \tag{135}$$

Furthermore, there are values A and $B = B(f)$ such that for all $\sigma \ge 0$ and $n \ge 2$,

$$\mathbb{E}\big|\widehat{\mathrm{C}}_\infty(X_n,Y_n) - \mathrm{C}_\infty(f,g)\big| \le AL\sigma\sqrt{\frac{\log(n)}{n}} + B\max\left\{\frac{\log(n)}{n^r},\frac{1}{n^2}\right\} \tag{136}$$

2. *Concentration bound. For $1 \le p < \infty$, there is a value $C = C(p,f)$ such that for all $\sigma > 0$, $t \ge 0$, and $n \ge 2$,*

$$\mathbb{P}\left\{\big|\widehat{\mathrm{C}}_p(X_n,Y_n) - \mathrm{C}_p(f,g)\big| \ge t\right\} \le 2\exp\left(-C\frac{nt^2}{\sigma^2}\right). \tag{137}$$

Furthermore, there is a value $C = C(f)$ such that for all $\sigma > 0$, $t \ge 0$, and $n \ge 2$,

$$\mathbb{P}\left\{\big|\widehat{\mathrm{C}}_\infty(X_n,Y_n) - \mathrm{C}_\infty(f,g)\big| \ge t\right\} \le 2\exp\left(-C\frac{nt^2}{\sigma^2\log(n)}\right). \tag{138}$$

3. *Almost sure limit. For all $1 \le p \le \infty$, $\widehat{\mathrm{C}}_p(X_n,Y_n) \to \mathrm{C}_p(f,g)$ almost surely as $n \to \infty$.*

Remark 17. It is straightforward to extend the definition of $\widehat{\mathrm{C}}_p(x,y)$, and the results from Theorem 4.1, to the setting where x and y contain samples of f and g taken on different grids, by interpolating the estimated Volterra transforms onto a common grid.

Theorem 4.1 is an easy corollary of the following two results:

Proposition 4.2. *Suppose f is a C^r function on $\mathbb{R}$, $r > 1$, that is supported on $[a,b]$ and satisfies $\int_a^b f = 0$. Let $x \in \mathbb{R}^n$ have entries $x[j] = f(t_j)$, $0 \le j \le n-1$. Then for all $1 \le p \le \infty$, there is a constant $C = C(p,f)$ such that*

$$|\mathrm{V}_p(x) - \|f\|_{V^p}| \le C\max\left\{\frac{\log(n)}{n^r},\frac{1}{n^2}\right\} \tag{139}$$

for $n \ge 2$.

Proposition 4.3. *Let $Z[0], Z[1], \dots, Z[n], \dots,$ be independent, mean-zero sub-Gaussian random variables with sub-Gaussian norms $\sigma_j = \|Z[j]\|_{\psi_2}$; and suppose $\sigma > 0$ satisfies*

$$\frac{1}{n}\sum_{j=0}^{n-1}\sigma_j^2 \le \sigma^2, \tag{140}$$

for all n. For each $n \ge 2$, let $Z_n = (Z[0], \dots, Z[n-1])$. Then:

1. *Expectation bound. There is a universal constant $C > 0$ such that for all $1 \le p < \infty$, $\sigma \ge 0$, and $n \ge 2$,*

$$\mathbb{E}\left[\mathrm{V}_p(Z_n)\right] \le CL^{1+1/p}\sqrt{p}\,\sigma\frac{1}{\sqrt{n}}, \tag{141}$$

and

$$\mathbb{E}\left[\mathrm{V}_\infty(Z_n)\right] \le CL\sigma\sqrt{\frac{\log(n)}{n}}. \tag{142}$$

2. *Concentration bound. For $1 \le p < \infty$, there is a value $C = C(p)$ such that for all $\sigma > 0$, $t \ge 0$, and $n \ge 2$,*

$$\mathbb{P}\left\{\mathrm{V}_p(Z_n) \ge t\right\} \le 2\exp\left(-C\frac{nt^2}{L^{2+2/p}\sigma^2}\right). \tag{143}$$

Furthermore, there is a universal constant $C > 0$ such that for all $\sigma > 0$, $t \ge 0$, and $n \ge 2$,

$$\mathbb{P}\left\{\mathrm{V}_\infty(Z_n) \ge t\right\} \le 2\exp\left(-C\frac{nt^2}{L^2\sigma^2\log(n)}\right) \tag{144}$$

for $2 < p \le \infty$.

3. Almost sure limit. For all $1 \le p \le \infty$, $\mathrm{V}_p(Z_n) \to 0$ almost surely as $n \to \infty$.

Remark 18. When p is an even integer, the bound in Proposition 4.2 can be sharpened to $O(\log(n)/n^r)$.

We now show that Theorem 4.1 follows from Propositions 4.2 and 4.3:

Proof of Theorem 4.1. It is enough to show the analogous results for the Volterra norm of f alone; we can then replace f by $f - g$. Let $x[j] = f(t_j)$, $0 \le j \le n-1$, so that $X_n = x + Z_n$. We have

$$\mathrm{V}_p(X_n) - \|f\|_{V^p} = \mathrm{V}_p(x + Z_n) - \|f\|_{V^p} \le \mathrm{V}_p(x) - \|f\|_{V^p} + \mathrm{V}_p(Z_n), \tag{145}$$

and

$$\mathrm{V}_p(X_n) - \|f\|_{V^p} = \mathrm{V}_p(x + Z_n) - \|f\|_{V^p} \ge \mathrm{V}_p(x) - \|f\|_{V^p} - \mathrm{V}_p(Z_n), \tag{146}$$

and therefore

$$|\mathrm{V}_p(X_n) - \|f\|_{V^p}| \le |\mathrm{V}_p(x) - \|f\|_{V^p}| + \mathrm{V}_p(Z_n). \tag{147}$$

The expected error bounds (135) and (136) are then immediate, as is almost sure convergence. To show concentration: suppose first that $1 \le p < \infty$. Then for all $t \ge 0$ and $n \ge 2$,

$$\mathbb{P}\left\{\mathrm{V}_p(Z_n) \ge t\right\} \le 2\exp\left(-C\frac{nt^2}{\sigma^2}\right), \tag{148}$$

where C may depend on p and L. By Proposition 4.2, by making C smaller, we can ensure that for all $n \ge 2$,

$$|\mathrm{V}(x) - \|f\|_{V^p}| \le \sqrt{\frac{\log(2)\sigma^2}{Cn}}, \tag{149}$$

and therefore, setting $\eta = \log(2)/C$,

$$|\mathrm{V}_p(X_n) - \|f\|_{V^p}| \le \sqrt{\eta\frac{\sigma^2}{n}} + \mathrm{V}_p(Z_n). \tag{150}$$

Now, given $t \ge 0$, we consider two cases. First, if n is large enough so that $\sqrt{\eta\,\sigma^2/n} \le t/2$, then

$$\begin{aligned}\mathbb{P}\left\{|\mathrm{V}_p(X_n) - \|f\|_{V^p}| \ge t\right\} &\le \mathbb{P}\left\{\mathrm{V}_p(Z_n) \ge t - \sqrt{\eta\,\sigma^2/n}\right\} \\ &\le \mathbb{P}\left\{\mathrm{V}_p(Z_n) \ge t/2\right\} \\ &\le 2\exp\left\{-Cn(t/2)^2/\sigma^2\right\} \\ &= 2\exp\left\{-(C/4)nt^2/\sigma^2\right\}\end{aligned} \tag{151}$$

On the other hand, if $\sqrt{\eta\,\sigma^2/n} \ge t/2$, then this same bound holds trivially, since

$$2\exp\left(-(C/4)\frac{nt^2}{\sigma^2}\right) = 2\exp\left(-C\frac{n(t/2)^2}{\sigma^2}\right) \ge 2\exp\left(-C\frac{n(\eta\,\sigma^2/n)}{\sigma^2}\right) = 2\exp\left(-C\eta\right) = 1, \tag{152}$$

which exceeds $\mathbb{P}\left\{|\mathrm{V}_p(X_n) - \|f\|_{V^p}| \ge t\right\}$. So, for all $n \ge 2$ and all $t \ge 0$, the bound

$$\mathbb{P}\left\{|\mathrm{V}_p(X_n) - \|f\|_{V^p}| \ge t\right\} \le 2\exp\left(-(C/4)\frac{nt^2}{\sigma^2}\right) \tag{153}$$

holds, as desired. The proof for $p = \infty$ is similar.

□

We now turn to the proofs of Propositions 4.2 and 4.3.

4.1.1 Proof of Proposition 4.2

Lemma 4.4. *Let f satisfy the conditions of Proposition 4.2, and let α be defined as in* (127)*. Then there is a value $C > 0$ such that for all even $n \geq 2$ and $|k| < n/2$,*

$$\left|\alpha[k] - \widehat{f}(k/L)\right| \leq \frac{C}{n^r}. \tag{154}$$

Proof. This is an immediate consequence of Lemma 6.1.

□

Corollary 4.5. *Let f satisfy the conditions of Proposition 4.2, and let ν_x be defined as in* (130)*. Then there is a value $C > 0$ such that for $|t| < L$ and $n \geq 2$,*

$$|\nu_x(t) - (\mathcal{V}f)(t)| \leq C\frac{\log(n)}{n^r}. \tag{155}$$

Proof. Applying Lemma 4.4 we have:

$$\left|\sum_{0<|k|<n/2} \beta[k]e^{2\pi \mathrm{i}kt/L} - \sum_{0<|k|<n/2} \widehat{(\mathcal{V}f)}(k/L)e^{2\pi \mathrm{i}kt/L}\right| \leq C \sum_{0<|k|<n/2} \frac{\left|\alpha[k] - \widehat{f}(k/L)\right|}{|k|} \leq C\frac{\log(n)}{n^r}. \tag{156}$$

Since f is C^r, $|\widehat{f}(k/L)| = O(|k|^{-r})$, and therefore $|\widehat{(\mathcal{V}f)}(k/L)| = O(|k|^{-(r+1)})$, and so the tail may be bounded

$$\left|\sum_{|k|\geq n/2} \widehat{(\mathcal{V}f)}(k/L)e^{2\pi \mathrm{i}kt/L}\right| \leq C \sum_{|k|\geq n/2} \frac{1}{|k|^{r+1}} \leq \frac{C}{n^r}. \tag{157}$$

Therefore,

$$\begin{aligned}\left|\sum_{0<|k|<n/2} \beta[k]e^{2\pi \mathrm{i}kt/L} - \sum_{k\neq 0} \widehat{(\mathcal{V}f)}(k/L)e^{2\pi \mathrm{i}kt/L}\right| &\leq \left|\sum_{0<|k|<n/2} (\beta[k] - \widehat{(\mathcal{V}f)}(k/L))e^{2\pi \mathrm{i}kt/L}\right| + \left|\sum_{|k|\geq n/2} \widehat{(\mathcal{V}f)}(k/L)e^{2\pi \mathrm{i}kt/L}\right| \\ &\leq \sum_{0<|k|<n/2} \left|\beta[k] - \widehat{(\mathcal{V}f)}(k/L))\right| + \sum_{|k|\geq n/2} \left|\widehat{(\mathcal{V}f)}(k/L)\right| \\ &\leq C\frac{\log(n)}{n^r}. \end{aligned} \tag{158}$$

Taking $t = a$ also shows

$$\left|\beta[0] - \widehat{(\mathcal{V}f)}(0)\right| = \left|\sum_{0<|k|<n/2} \beta[k]e^{2\pi \mathrm{i}ka/L} - \sum_{k\neq 0} \widehat{(\mathcal{V}f)}(k/L)e^{2\pi \mathrm{i}ka/L}\right| \leq C\frac{\log(n)}{n^r}, \tag{159}$$

and consequently

$$|\nu_x(t) - (\mathcal{V}f)(t)| \leq \left|\beta[0] - \widehat{(\mathcal{V}f)}(0)\right| + \left|\sum_{0<|k|<n/2} \beta[k]e^{2\pi \mathrm{i}ks/L} - \sum_{k\neq 0} \widehat{(\mathcal{V}f)}(k/L)e^{2\pi \mathrm{i}ks/L}\right| \leq C\frac{\log(n)}{n^r}. \tag{160}$$

□

We now complete the proof of Proposition 4.2. When $1 \leq p < \infty$, from the triangle inequality

$$\begin{aligned}
\left|\mathrm{V}_p(x) - \|f\|_{V^p}\right| &= \left| \left(\frac{L}{n} \sum_{j=0}^{n-1} |\nu_x(t_j)|^p \right)^{1/p} - \|f\|_{V^p} \right| \\
&\leq \left| \left(\frac{L}{n} \sum_{j=0}^{n-1} |\nu_x(t_j)|^p \right)^{1/p} - \left(\frac{L}{n} \sum_{j=0}^{n-1} |(\mathcal{V}f)(t_j)|^p \right)^{1/p} \right| + \left| \left(\frac{L}{n} \sum_{j=0}^{n-1} |(\mathcal{V}f)(t_j)|^p \right)^{1/p} - \|f\|_{V^p} \right| \\
&\leq \left(\frac{L}{n} \sum_{j=0}^{n-1} |\nu_x(t_j) - (\mathcal{V}f)(t_j)|^p \right)^{1/p} + \left| \left(\frac{L}{n} \sum_{j=0}^{n-1} |(\mathcal{V}f)(t_j)|^p \right)^{1/p} - \|\mathcal{V}f\|_{L^p} \right|, \qquad (161)
\end{aligned}$$

with obvious modifications when $p = \infty$. From Corollary 4.5, the first term is $O(\log(n)/n^r)$, and from Corollary 6.5, the second term is $O(1/n^2)$. We have therefore shown that for all n sufficiently large,

$$\left|\mathrm{V}_p(x) - \|f\|_{V^p}\right| \leq C \max\left\{ \frac{\log(n)}{n^r}, \frac{1}{n^2} \right\}, \qquad (162)$$

completing the proof of Proposition 4.2.

4.1.2 Proof of Proposition 4.3

First, we note that the third part of the Proposition (almost sure convergence) follows immediately from the second part (the concentration bound) from Lemma 6.7.

Now, recall that the vectors α and β are defined as follows:

$$\alpha[k] = \frac{L}{n} \sum_{j=0}^{n-1} Z[j] e^{-2\pi \mathrm{i} k t_j / L}, \quad 0 \leq |k| < n/2, \qquad (163)$$

$$\beta[k] = \frac{\alpha[k]}{2\pi \mathrm{i} k / L}, \quad 0 < |k| < n/2, \qquad (164)$$

and when $k = 0$,

$$\beta[0] = -\sum_{0<|k|<n/2} \beta[k] e^{2\pi \mathrm{i} k a / L}. \qquad (165)$$

Define the random vector W by $W[j] = \nu_Z(t_j)$, that is,

$$W[j] = \frac{1}{L} \sum_{k=-n/2+1}^{n/2-1} \beta[k] e^{2\pi \mathrm{i} t_j k / L}, \qquad (166)$$

for $0 \leq j \leq n-1$. Then for $1 \leq p < \infty$,

$$\mathrm{V}_p(Z_n) = \left(\frac{L}{n} \sum_{j=0}^{n-1} |W[j]|^p \right)^{1/p}, \qquad (167)$$

and $\mathrm{V}_\infty(Z_n) = \|W\|_\infty$.

It will be convenient to define the auxiliary vector X by

$$X[j] = \sum_{0<|k|<n/2} \beta[k] e^{2\pi \mathrm{i} t_j k / L}. \qquad (168)$$

Then $W[j] = (X[j] + \beta[0])/L$.

Expectation of $\mathrm{V}_p(Z_n)$. For a fixed $0 \le \ell \le n-1$,

$$\begin{aligned}
X[\ell] &= \sum_{0<|k|<n/2} \beta[k] e^{2\pi \mathrm{i} t_\ell k/L} \\
&= \frac{L}{2\pi\mathrm{i}} \sum_{0<|k|<n/2} \frac{\alpha[k]}{k} e^{2\pi \mathrm{i} t_\ell k/L} \\
&= \frac{L}{2\pi\mathrm{i}} \sum_{0<|k|<n/2} \frac{1}{k} \left(\frac{L}{n} \sum_{j=0}^{n-1} Z[j] e^{-2\pi \mathrm{i} k t_j/L} \right) e^{2\pi \mathrm{i} t_\ell k/L} \\
&= \frac{L^2}{2\pi \mathrm{i} n} \sum_{j=0}^{n-1} Z[j] \sum_{0<|k|<n/2} \frac{e^{2\pi \mathrm{i} k(\ell-j)/n}}{k} \\
&= \frac{L^2}{\pi n} \sum_{j=0}^{n-1} Z[j] \sum_{k=1}^{n/2-1} \frac{\sin(2\pi(\ell-j)k/n)}{k} \\
&= \frac{L^2}{\pi n} \sum_{j=0}^{n-1} Q_n[\ell-j] Z[j],
\end{aligned} \tag{169}$$

where

$$Q_n[m] = \sum_{k=1}^{n/2-1} \frac{\sin(2\pi m k/n)}{k}. \tag{170}$$

By Lemma 6.6, $|Q_n[m]| \le C$, for all m and an absolute constant C. Since the $Z[j]$ are independent and mean zero, we may apply Proposition 6.12 to get the bound

$$\|X[\ell]\|_{\psi_2}^2 \le C \frac{L^4}{n^2} \sum_{j=0}^{n-1} Q_n[\ell-j]^2 \|Z[j]\|_{\psi^2}^2 \le C L^4 \frac{\sigma^2}{n}. \tag{171}$$

A nearly identical proof shows that $\|\beta[0]\|_{\psi_2}^2 \le C L^4 \sigma^2/n$. Since $W[j] = (X[j] + \beta[0])/L$, it follows that, for all $0 \le j \le n-1$,

$$\|W[j]\|_{\psi_2} \le C L \frac{\sigma}{\sqrt{n}}, \tag{172}$$

and hence, for any $1 \le p < \infty$, from Proposition 6.8,

$$\mathbb{E}[|W[j]|^p] \le \left(C\sqrt{p} L \frac{\sigma}{\sqrt{n}} \right)^p. \tag{173}$$

Consequently,

$$\mathbb{E}[\mathrm{V}_p(Z_n)^p] = \frac{L}{n} \sum_{j=0}^{n-1} \mathbb{E}[|W[j]|^p] \le \left(C\sqrt{p} L^{1+1/p} \frac{\sigma}{\sqrt{n}} \right)^p, \tag{174}$$

and therefore, using Jensen's inequality,

$$\mathbb{E}[\mathrm{V}_p(Z_n)] \le \left(\mathbb{E}[\mathrm{V}_p(Z_n)^p]\right)^{1/p} \le C\sqrt{p} L^{1+1/p} \frac{\sigma}{\sqrt{n}}. \tag{175}$$

The case for $p = \infty$ follows immediately from Lemma 6.10.

Concentration of $\mathrm{V}_p(Z_n)$. Fix $1 \le p < \infty$, so that

$$\mathrm{V}_p(Z_n)^p = \frac{L}{n} \sum_{j=0}^{n-1} |W[j]|^p. \tag{176}$$

Since each $\|W[j]\|_{\psi_2} \leq CL\sigma/\sqrt{n}$, therefore $|W[j]|^p$ is sub-Weibull$(2/p)$ (see Section 6.2.2), with

$$\||W[j]|^p\|_{\psi_{2/p}} \leq C^p L^p \sigma^p / n^{p/2}, \tag{177}$$

and so, by Proposition 6.13,

$$\|\mathrm{V}_p(Z_n)^p\|_{\psi_{2/p}} \leq C_p L^{p+1} \sigma^p / n^{p/2}. \tag{178}$$

Therefore, $\|\mathrm{V}_p(Z_n)\|_{\psi_2} = C_p L^{1+1/p}\sigma/\sqrt{n}$, and hence, for all $t \geq 0$,

$$\mathbb{P}\{\mathrm{V}_p(Z_n) \geq t\} \leq 2\exp\left\{-C_p \frac{nt^2}{L^{2+2/p}\sigma^2}\right\}, \tag{179}$$

as claimed.

When $p = \infty$, we may apply Proposition 6.11 to $\mathrm{V}_\infty(Z_n) = \max_{0\leq j\leq n-1} |W[j]|$ to obtain the bound

$$\|\mathrm{V}_\infty(Z_n)\|_{\psi_2} \leq C\sqrt{\log(n)} \max_{1\leq j\leq n} \|W[j]\|_{\psi_2} \leq CL\sigma\sqrt{\frac{\log(n)}{n}}. \tag{180}$$

Consequently, for any $t \geq 0$,

$$\mathbb{P}\{\mathrm{V}_\infty(Z_n) \geq t\} \leq 2\exp\left\{-C\frac{nt^2}{L^2\sigma^2\log(n)}\right\}, \tag{181}$$

which is the claimed bound.

4.2 Robustness to noise in 2D

We turn now to the discrete approximation of the sliced Cramér distance in 2D, with respect to the uniform measure over $\mathbb{S}^1$. We use the discretization described in [65] apply the non-uniform discrete Fourier transform to compute the Radon transform of the input functions. Let $R > 0$ and $L = 2R$. Let n a positive integer; throughout this discussion, we will assume for simplicity that n is even. Let $x \in \mathbb{R}^{n^2} = \mathbb{R}^n \times \mathbb{R}^n$; the reader should think of the entries $x[i,j]$ of x as (possibly noisy) samples from a function $f : \mathbb{R}^2 \to \mathbb{R}$ supported on the disc of radius R centered at the origin, that is, $x[i,j] \approx f(t_i, t_j)$, $0 \leq i,j \leq n-1$, where $t_j = -R + 2Rj/n$. Throughout the section, we will denote by $\mathcal{P}_\theta$ the tomographic projection operator onto $(\cos(\theta), \sin(\theta))$.

For $\theta \in [0,\pi)$, define the values

$$\alpha_\theta[k] = \frac{L^2}{n^2}\sum_{i,j} x[i,j] e^{-2\pi \mathrm{i} k(t_i\cos(\theta)+t_j\sin(\theta))/L}. \tag{182}$$

Then for $0 \leq |k| < n/2$, $\alpha_\theta[k] \approx \widehat{f}((k/L)\cos(\theta), (k/L)\sin(\theta)) = \widehat{(\mathcal{P}_\theta f)}(k/L)$.

For $0 < |k| < n/2$, let

$$\beta_\theta[k] = \frac{\alpha_\theta[k]}{2\pi\mathrm{i}k/L}. \tag{183}$$

For $k = 0$, define

$$\beta_\theta[0] = -\sum_{0<|k|<n/2} \beta_\theta[k] e^{-2\pi\mathrm{i}kR/L} = -\sum_{0<|k|<n/2} \beta_\theta[k](-1)^k. \tag{184}$$

Define $\nu_x(t,\theta)$ by

$$\nu_x(t,\theta) = \frac{1}{L}\sum_{k=-n/2+1}^{n/2-1} \beta_\theta[k] e^{2\pi\mathrm{i}tk/L}. \tag{185}$$

Then $\nu_x(t,\theta) \approx (\mathcal{V}\mathcal{P}_\theta f)(t)$.

Let $\theta_\ell = \pi\ell/n$, $\ell = 0, \ldots, n-1$. We define the estimated sliced Volterra norm for $1 \le p < \infty$ to be

$$\mathrm{SV}_p(x) = \left(\frac{1}{n} \sum_{\ell=0}^{n-1} \frac{L}{n} \sum_{j=0}^{n} |\nu_x(t_j, \theta_\ell)|^p \right)^{1/p}, \tag{186}$$

and

$$\mathrm{SV}_\infty(x) = \max_{0 \le j,\ell \le n-1} |\nu_x(t_j, \theta_\ell)|. \tag{187}$$

Given two vectors x and y, define their sliced Cramér distance to be

$$\widehat{\mathrm{SC}}_p(x, y) = \mathrm{SV}_p(x - y). \tag{188}$$

Remark 19. Using a non-uniform Fast Fourier Transform (NUFFT) (for example, see [16, 17, 18, 24, 4, 5]) to evaluate the $\alpha_{\theta_\ell}[k]$ and the $\nu_x(t_j, \theta_\ell)$, the entire computation described here can be performed at cost $O(n^2 \log n)$. In our implementation, we use the Flatiron Institute NUFFT (FINUFFT) [4, 5].

We can now state the main result from this section, which says that the discrete sliced Cramér distance in 2D is robust to additive heteroscedastic sub-Gaussian noise:

Theorem 4.6. *Suppose f and g are C^r functions on $\mathbb{R}^2$, $r > 2$, that are supported on the disc of radius $R > 0$ centered at the origin, and which satisfy $\int_{\mathbb{R}^2} f = \int_{\mathbb{R}^2} g$. Let $Z[j,k]$, $\widetilde{Z}[j,k]$, $0 \le j, k \le n-1$, be independent, mean-zero sub-Gaussian random variables with sub-Gaussian norms $\sigma_{jk} = \|Z[j,k]\|_{\psi_2}$ and $\widetilde{\sigma}_{jk} = \|\widetilde{Z}[j,k]\|_{\psi_2}$; and suppose that for all n, $\sigma > 0$ satisfies*

$$\frac{1}{n^2} \sum_{k=0}^{n-1} \sum_{j=0}^{n-1} \sigma_{jk}^2 + \frac{1}{n^2} \sum_{k=0}^{n-1} \sum_{j=0}^{n-1} \widetilde{\sigma}_{jk}^2 \le \sigma^2. \tag{189}$$

Let X_n and Y_n be vectors in $\mathbb{R}^{n^2}$ with entries $X_n[j,k] = f(t_j, t_k) + Z[j,k]$ and $Y_n[j,k] = g(t_j, t_k) + \widetilde{Z}[j,k]$. Then:

1. *Expected error. For $1 \le p < \infty$, there are values A and $B = B(p, f)$ such that for all $\sigma \ge 0$ and $n \ge 2$,*

$$\mathbb{E}\left|\widehat{\mathrm{SC}}_p(X_n, Y_n) - \mathrm{SC}_p(f, g)\right| \le AL^{2+1/p}\sqrt{p}\frac{\sigma}{n} + \frac{B}{n^2}. \tag{190}$$

   Furthermore, there are values A and $B = B(f)$ such that for all $\sigma \ge 0$ and $n \ge 2$,

$$\mathbb{E}\left|\widehat{\mathrm{SC}}_\infty(X_n, Y_n) - \mathrm{SC}_\infty(f, g)\right| \le AL^2\sigma\frac{\sqrt{\log(n)}}{n} + \frac{B}{n^2}. \tag{191}$$

2. *Concentration bound. For $1 \le p < \infty$, there is a value $C = C(p, f)$ such that for all $\sigma > 0$, $t \ge 0$, and $n \ge 2$,*

$$\mathbb{P}\left\{\left|\widehat{\mathrm{SC}}_p(X_n, Y_n) - \mathrm{SC}_p(f, g)\right| \ge t\right\} \le 2\exp\left(-C\frac{n^2 t^2}{\sigma^2}\right). \tag{192}$$

   Furthermore, there is a value $C = C(f)$ such that for all $\sigma > 0$, $t \ge 0$, and $n \ge 2$,

$$\mathbb{P}\left\{\left|\widehat{\mathrm{SC}}_\infty(X_n, Y_n) - \mathrm{SC}_\infty(f, g)\right| \ge t\right\} \le 2\exp\left(-C\frac{n^2 t^2}{\sigma^2 \log(n)}\right). \tag{193}$$

3. *Almost sure limit. For all $1 \le p \le \infty$, $\widehat{\mathrm{SC}}_p(X_n, Y_n) \to \mathrm{SC}_p(f, g)$ almost surely as $n \to \infty$.*

Theorem 4.6 is a corollary of the following two results (the proof is the same as in the 1D case):

Proposition 4.7. *Suppose f is a C^r function on $\mathbb{R}^2$, $r > 2$, that is supported on the disc of radius $R > 0$ centered at the origin, and $\int_{\mathbb{R}^2} f = 0$. Let $1 \le p \le \infty$, and let $x \in \mathbb{R}^{n^2}$ have entries $x[j,k] = f(t_j, t_k)$, $0 \le j, k \le n-1$. Then for all $1 \le p \le \infty$, there is a constant $C = C(p, f)$ such that*

$$|\mathrm{SV}_p(x) - \|f\|_{SV^p}| \le \frac{C}{n^2}, \tag{194}$$

for $n \ge 2$.

Proposition 4.8. *Let $Z[j,k]$, $j,k \geq 0$, be independent, mean-zero sub-Gaussian random variables with sub-Gaussian norms $\sigma_{jk} = \|Z[j,k]\|_{\psi_2}$; and suppose $\sigma > 0$ satisfies*

$$\frac{1}{n^2} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} \sigma_{jk}^2 \leq \sigma^2, \tag{195}$$

for all n. For each $n \geq 2$, let $Z_n \in \mathbb{R}^{n^2}$ have entries $Z[j,k]$. Then:

1. *Expectation bound. There is a universal constant $C > 0$ such that for all $1 \leq p < \infty$, $\sigma \geq 0$, and $n \geq 2$,*

$$\mathbb{E}\left[\mathrm{SV}_p(Z_n)\right] \leq C L^{2+1/p} \sqrt{p}\, \frac{\sigma}{n}, \tag{196}$$

and

$$\mathbb{E}\left[\mathrm{SV}_\infty(Z_n)\right] \leq C L^2 \sigma \frac{\sqrt{\log(n)}}{n}. \tag{197}$$

2. *Concentration bound. For $1 \leq p < \infty$, there is a value $C = C(p)$ such that for all $\sigma > 0$, $t \geq 0$, and $n \geq 2$,*

$$\mathbb{P}\left\{\mathrm{SV}_p(Z_n) \geq t\right\} \leq 2 \exp\left(-C \frac{n^2 t^2}{L^{4+2/p} \sigma^2}\right). \tag{198}$$

Furthermore, there is a universal constant $C > 0$ such that for all $\sigma > 0$, $t \geq 0$, and $n \geq 2$,

$$\mathbb{P}\left\{\mathrm{SV}_\infty(Z_n) \geq t\right\} \leq 2 \exp\left(-C \frac{n^2 t^2}{L^4 \sigma^2 \log(n)}\right) \tag{199}$$

for $2 < p \leq \infty$.

3. *Almost sure limit. For all $1 \leq p \leq \infty$, $\mathrm{SV}_p(Z_n) \to 0$ almost surely as $n \to \infty$.*

4.2.1 Proof of Proposition 4.7

Lemma 4.9. *Let f satisfy the conditions of Proposition 4.7, and let α_θ be defined as in (182). Then there is a $C > 0$ such that*

$$\left|\alpha_\theta[m] - \widehat{(\mathcal{P}_\theta f)}(m/L)\right| \leq \frac{C}{n^r} \tag{200}$$

for all $0 \leq \theta < \pi$, $n \geq 2$, and $|m| < n/2$.

Proof. This is an immediate consequence of Lemma 6.1. ∎

Corollary 4.10. *Let f satisfy the conditions of Proposition 4.7, and let ν_x be defined as in (185). Then there is a $C > 0$ such that*

$$|\nu_x(t,\theta) - (\mathcal{V}\mathcal{P}_\theta f)(t)| \leq C \frac{\log(n)}{n^r} \tag{201}$$

for all $0 \leq \theta < \pi$ and $|t| < L$.

Proof. For all t and θ, applying Lemma 4.9 gives

$$\left| \frac{1}{L} \sum_{0<|k|\leq n/2-1} \beta_\theta[k] e^{2\pi \mathrm{i} tk/L} - \frac{1}{L} \sum_{0<|k|\leq n/2-1} \frac{\widehat{(\mathcal{P}_\theta f)}(k/L)}{2\pi \mathrm{i} k/L} e^{2\pi \mathrm{i} tk/L} \right| \leq \frac{C}{n^r} \sum_{0<|k|\leq n/2-1} \frac{1}{|k|} \leq C \frac{\log(n)}{n^r}, \tag{202}$$

and the tail can be bounded

$$\left| \sum_{|k|\geq n/2} \frac{\widehat{(\mathcal{P}_\theta f)}(k/L)}{2\pi \mathrm{i} k} e^{2\pi \mathrm{i} tk/L} \right| = \left| \sum_{|k|\geq n/2} \frac{\widehat{f}((k/L)\cos(\theta), (k/L)\sin(\theta))}{2\pi \mathrm{i} k} e^{2\pi \mathrm{i} tk/L} \right| \leq C \sum_{|k|\geq n/2} \frac{1}{k^{r+1}} \leq C \frac{1}{n^r}. \tag{203}$$

Similarly, for all θ,

$$\left|\beta_\theta[0] - (\widehat{\mathcal{V}\mathcal{P}_\theta f})(0)\right| \leq \sum_{0<|k|\leq n/2-1} \left|\beta_\theta[k] - \frac{\widehat{(\mathcal{P}_\theta f)}(k/L)}{2\pi \mathrm{i} k/L}\right| + \sum_{|k|\geq n/2} \left|\frac{\widehat{(\mathcal{P}_\theta f)}(k/L)}{2\pi \mathrm{i} k/L}\right| \leq C\frac{\log(n)}{n^r}. \tag{204}$$

It then follows that for all t and θ,

$$|\nu_x(t,\theta) - (\mathcal{V}\mathcal{P}_\theta f)(t)| \leq C\frac{\log(n)}{n^r}, \tag{205}$$

where C does not depend on θ, t or n.

□

For brevity, let $G(y,\theta) = (\mathcal{V}\mathcal{P}_\theta f)(y)$, i.e.

$$G(y,\theta) = \int_{-R}^{y}\int_{-R}^{R} f(s\cos(\theta) + t\sin(\theta), t\cos(\theta) - s\sin(\theta))\, dt\, ds. \tag{206}$$

Then from Corollary 4.10,

$$\begin{aligned}\left|\mathrm{SV}_p(x) - \left(\frac{1}{n}\sum_{\ell=0}^{n-1}\frac{L}{n}\sum_{j=0}^{n-1}|G(t_j,\theta_\ell)|^p\right)^{1/p}\right| &= \left|\left(\frac{1}{n}\sum_{\ell=0}^{n-1}\frac{L}{n}\sum_{j=0}^{n-1}|\nu_x(t_j,\theta_\ell)|^p\right)^{1/p} - \left(\frac{1}{n}\sum_{\ell=0}^{n-1}\frac{L}{n}\sum_{j=0}^{n-1}|G(t_j,\theta_\ell)|^p\right)^{1/p}\right| \\ &\leq \left(\frac{1}{n}\sum_{\ell=0}^{n-1}\frac{L}{n}\sum_{j=0}^{n-1}|\nu_x(t_j,\theta_\ell) - G(t_j,\theta_\ell)|^p\right)^{1/p} \\ &\leq C\frac{\log(n)}{n^r},\end{aligned} \tag{207}$$

with the obvious modifications when $p = \infty$.

Define the function $I_p(\theta) = \|G(\cdot,\theta)\|_{L^p(dt)}$. We then have the following lemma:

Lemma 4.11. *Let f satisfy the conditions of Proposition 4.7. For all $1 \leq p \leq \infty$, there is an N_p such that for all $n \geq N_p$,*

$$\left|\left(\frac{1}{n}\sum_{\ell=0}^{n-1} I_p(\theta_\ell)^p\right)^{1/p} - \|f\|_{SV^p}\right| \leq CL^{2+1/p}\left\|\frac{\partial^2 G}{\partial\theta^2}\right\|_{L^\infty}\frac{1}{n^2}, \tag{208}$$

with the obvious modification when $p = \infty$, where C is a universal constant.

Proof. First suppose $1 \leq p < \infty$. For each $k \neq 0$,

$$\int_0^{2\pi} I_p(\theta)^p e^{-\mathrm{i}k\theta}\, d\theta = \int_0^{2\pi}\left(\int_{-R}^{R}|G(t,\theta)|^p\, dt\right)e^{-\mathrm{i}k\theta}\, d\theta = \int_{-R}^{R}\int_0^{2\pi}|G(t,\theta)|^p e^{-\mathrm{i}k\theta}\, d\theta\, dt, \tag{209}$$

and by Lemma 6.3, for all t we have the bound

$$\left|\int_0^{2\pi}|G(t,\theta)|^p e^{-\mathrm{i}k\theta}\, d\theta\right| \leq C\frac{L^2 p}{k^2}\left\|\frac{\partial^2 G}{\partial\theta^2}\right\|_{L^\infty}\int_0^{2\pi}|G(t,\theta)|^{p-1}d\theta, \tag{210}$$

where C is universal; therefore,

$$\left|\int_0^{2\pi} I_p(\theta)^p e^{-\mathrm{i}k\theta}\, d\theta\right| \leq C\frac{L^2 p}{k^2}\left\|\frac{\partial^2 G}{\partial\theta^2}\right\|_{L^\infty}\int_{-R}^{R}\int_0^{2\pi}|G(t,\theta)|^{p-1}\, d\theta\, dt. \tag{211}$$

By Lemma 6.2, it then follows that, for all $1 \leq p < \infty$,

$$\left|\frac{1}{2n}\sum_{\ell=0}^{2n-1} I_p(2\pi\ell/2n)^p - \frac{1}{2\pi}\int_0^{2\pi} I_p(\theta)^p\, d\theta\right| \leq C\frac{L^2 p}{n^2}\left\|\frac{\partial^2 G}{\partial\theta^2}\right\|_{L^\infty}\int_{-R}^{R}\int_0^{2\pi}|G(t,\theta)|^{p-1}\, d\theta\, dt, \tag{212}$$

and, since I_p is π-periodic, we have

$$\frac{1}{2n}\sum_{\ell=0}^{2n-1} I_p(2\pi\ell/2n)^p = \frac{1}{n}\sum_{\ell=0}^{n-1} I_p(\pi\ell/n)^p = \frac{1}{n}\sum_{\ell=0}^{n-1} I_p(\theta_\ell)^p, \tag{213}$$

so that

$$\begin{aligned}
\left|\frac{1}{n}\sum_{\ell=0}^{n-1} I_p(\theta_\ell)^p - \|f\|_{SV^p}^p\right| &= \left|\frac{1}{n}\sum_{\ell=0}^{n-1} I_p(\theta_\ell)^p - \frac{1}{2\pi}\int_0^{2\pi} I_p(\theta)^p\, d\theta\right| \\
&\le C\frac{L^2 p}{n^2}\left\|\frac{\partial^2 G}{\partial\theta^2}\right\|_{L^\infty}\int_{-R}^{R}\int_0^{2\pi}|G(t,\theta)|^{p-1}\, d\theta\, dt \\
&\le C\frac{pL^{2+1/p}}{n^2}\left\|\frac{\partial^2 G}{\partial\theta^2}\right\|_{L^\infty}\left(\int_{-R}^{R}\int_0^{2\pi}|G(t,\theta)|^{p}\, d\theta\, dt\right)^{(p-1)/p} \\
&= C\frac{pL^{2+1/p}}{n^2}\left\|\frac{\partial^2 G}{\partial\theta^2}\right\|_{L^\infty}\|f\|_{SV^p}^{p-1}.
\end{aligned}\tag{214}$$

Assuming $\|f\|_{SV^p}\neq 0$ (for otherwise $f\equiv 0$ and the proof is trivial), it follows that for all n sufficiently large,

$$\frac{1}{n}\sum_{\ell=0}^{n-1} I_p(\theta_\ell)^p \ge \frac{1}{2}\|f\|_{SV^p}^p, \tag{215}$$

and consequently, applying the mean value theorem to the function $y\mapsto y^{1/p}$,

$$\begin{aligned}
\left|\left(\frac{1}{n}\sum_{\ell=0}^{n-1} I_p(\theta_\ell)^p\right)^{1/p} - \|f\|_{SV^p}\right| &\le C\frac{\left(\|f\|_{SV^p}^p/2\right)^{1/p-1}}{p}\left(\frac{pL^{2+1/p}}{n^2}\left\|\frac{\partial^2 G}{\partial\theta^2}\right\|_{L^\infty}\|f\|_{SV^p}^{p-1}\right) \\
&\le CL^{2+1/p}\left\|\frac{\partial^2 G}{\partial\theta^2}\right\|_{L^\infty}\frac{1}{n^2},
\end{aligned}\tag{216}$$

as claimed.

When $p=\infty$: fix any t in $[-R,R]$. Then from Corollary 6.5

$$\left|\max_{0\le\ell\le n-1}|G(t,\theta_\ell)| - \max_{0\le\theta\le\pi}|G(t,\theta)|\right| \le CL^2\left\|\frac{\partial^2 G}{\partial\theta^2}\right\|_{L^\infty}\frac{1}{n^2}, \tag{217}$$

and so

$$\begin{aligned}
\left|\max_{0\le\ell\le n-1} I_\infty(\theta_\ell) - \|f\|_{SV^\infty}\right| &= \left|\max_{0\le\ell\le n-1}\max_{-R\le t\le R}|G(t,\theta_\ell)| - \max_{0\le\theta\le\pi}\max_{-R\le t\le R}|G(t,\theta)|\right| \\
&= \left|\max_{-R\le t\le R}\max_{0\le\ell\le n-1}|G(t,\theta_\ell)| - \max_{-R\le t\le R}\max_{0\le\theta\le\pi}|G(t,\theta)|\right| \\
&\le \max_{-R\le t\le R}\left|\max_{0\le\ell\le n-1}|G(t,\theta_\ell)| - \max_{0\le\theta\le\pi}|G(t,\theta)|\right| \\
&\le CL^2\left\|\frac{\partial^2 G}{\partial\theta^2}\right\|_{L^\infty}\frac{1}{n^2},
\end{aligned}\tag{218}$$

as claimed.

□

Lemma 4.12. *Let f satisfy the conditions of Proposition 4.7. For all $1\le p\le\infty$, there is an N_p such that for all $n\ge N_p$,*

$$\left|\left(\frac{2}{n}\sum_{\ell=0}^{n-1}\frac{L}{n}\sum_{j=0}^{n-1}|G(t_j,\theta_\ell)|^p\right)^{1/p} - \left(\frac{2}{n}\sum_{\ell=0}^{n-1}I_p(\theta_\ell)^p\right)^{1/p}\right| \le CL^{2+1/p}\left\|\frac{\partial^2 G}{\partial t^2}\right\|_{L^\infty}\frac{1}{n^2}, \tag{219}$$

with the obvious modification when $p=\infty$, where C is a universal constant.

Proof. First, suppose $1 \leq p < \infty$. By Corollary 6.4, for each θ,

$$\begin{aligned}\left|\frac{L}{n}\sum_{j=0}^{n-1}|G(t_j,\theta)|^p - I_p(\theta)^p\right| &= \left|\frac{L}{n}\sum_{j=0}^{n-1}|G(t_j,\theta)|^p - \int_{-R}^{R}|G(t,\theta)|^p dt\right| \\ &\leq C\frac{L^2 p}{n^2}\left\|\frac{\partial^2 G}{\partial t^2}\right\|_{L^\infty}\int_{-R}^{R}|G(t,\theta)|^{p-1}dt, \end{aligned} \tag{220}$$

and therefore,

$$\begin{aligned}\left|\frac{1}{n}\sum_{\ell=0}^{n-1}\frac{L}{n}\sum_{j=0}^{n-1}|G(t_j,\theta_\ell)|^p - \frac{1}{n}\sum_{\ell=0}^{n-1}I_p(\theta_\ell)^p\right| &\leq C\frac{L^2 p}{n^2}\left\|\frac{\partial^2 G}{\partial t^2}\right\|_{L^\infty}\left(\frac{1}{n}\sum_{\ell=0}^{n-1}\int_{-R}^{R}|G(t,\theta)|^{p-1}dt\right) \\ &\leq C\frac{L^{2+1/p}p}{n^2}\left\|\frac{\partial^2 G}{\partial t^2}\right\|_{L^\infty}\left(\frac{1}{n}\sum_{\ell=0}^{n-1}\int_{-R}^{R}|G(t,\theta)|^{p}dt\right)^{(p-1)/p} \\ &= C\frac{L^{2+1/p}p}{n^2}\left\|\frac{\partial^2 G}{\partial t^2}\right\|_{L^\infty}\left(\frac{1}{n}\sum_{\ell=0}^{n-1}I_p(\theta_\ell)^p\right)^{(p-1)/p}. \end{aligned} \tag{221}$$

By Lemma 4.11, for n sufficiently large,

$$\frac{1}{2^{1/p}}\|f\|_{SV^p} \leq \left(\frac{1}{n}\sum_{\ell=0}^{n-1}I_p(\theta_\ell)^p\right)^{1/p} \leq 2^{1/p}\|f\|_{SV^p}, \tag{222}$$

and so if n is also large enough to satisfy

$$n^2 \geq 2^{1+1/p}CL^{2+1/p}p\left\|\frac{\partial^2 G}{\partial t^2}\right\|_{L^\infty}\|f\|_{SV^p}^{-1} \tag{223}$$

then it is also true that

$$n^2 \geq 2CL^{2+1/p}p\left\|\frac{\partial^2 G}{\partial t^2}\right\|_{L^\infty}\left(\frac{1}{n}\sum_{\ell=0}^{n-1}I_p(\theta_\ell)^p\right)^{-1/p} \tag{224}$$

and consequently, for all sufficiently large n,

$$CL^{2+1/p}p\left\|\frac{\partial^2 G}{\partial t^2}\right\|_{L^\infty}\left(\frac{1}{n}\sum_{\ell=0}^{n-1}I_p(\theta_\ell)^p\right)^{(p-1)/p}\frac{1}{n^2} \leq \frac{1}{2}\left(\frac{1}{n}\sum_{\ell=0}^{n-1}I_p(\theta_\ell)^p\right), \tag{225}$$

in which case (221) implies

$$\frac{1}{n}\sum_{\ell=0}^{n-1}\frac{L}{n}\sum_{j=0}^{n-1}|G(t_j,\theta_\ell)|^p \geq \frac{1}{2}\left(\frac{1}{n}\sum_{\ell=0}^{n-1}I_p(\theta_\ell)^p\right) \geq \frac{1}{4}\|f\|_{SV^p}^p. \tag{226}$$

Since

$$\frac{1}{n}\sum_{\ell=0}^{n-1}I_p(\theta_\ell)^p \geq \frac{1}{2}\|f\|_{SV^p}^p \geq \frac{1}{4}\|f\|_{SV^p}^p, \tag{227}$$

by the mean value theorem applied to $y \mapsto y^{1/p}$ it follows that

$$\begin{aligned}
&\left| \left(\frac{1}{n} \sum_{\ell=0}^{n-1} \frac{L}{n} \sum_{j=0}^{n-1} |G(t_j, \theta_\ell)|^p \right)^{1/p} - \left(\frac{1}{n} \sum_{\ell=0}^{n-1} I_p(\theta_\ell)^p \right)^{1/p} \right| \\
&\leq \; C \frac{(\|f\|_{SV^p}^p/4)^{1/p-1}}{p} \frac{L^{2+1/p} p}{n^2} \left\| \frac{\partial^2 G}{\partial t^2} \right\|_{L^\infty} \left(\frac{2}{n} \sum_{\ell=0}^{n-1} I_p(\theta_\ell)^p \right)^{(p-1)/p} \\
&\leq \; C \frac{(\|f\|_{SV^p}^p/4)^{1/p-1}}{p} \frac{L^{2+1/p} p}{n^2} \left\| \frac{\partial^2 G}{\partial t^2} \right\|_{L^\infty} \left(2^{1/p} \|f\|_{SV^p} \right)^{(p-1)} \\
&\leq \; C L^{2+1/p} \left\| \frac{\partial^2 G}{\partial t^2} \right\|_{L^\infty} \frac{1}{n^2}. \qquad (228)
\end{aligned}$$

This completes the proof when $1 \leq p < \infty$.

When $p = \infty$: fix θ in $[0, \pi]$. By Corollary 6.5,

$$\left| \max_{0 \leq j \leq n-1} |G(t_j, \theta)| - I_\infty(\theta) \right| = \left| \max_{0 \leq j \leq n-1} |G(t_j, \theta)| - \max_{-R \leq t \leq R} |G(t, \theta)| \right| \leq C L^2 \left\| \frac{\partial^2 G}{\partial t^2} \right\|_{L^\infty} \frac{1}{n^2}, \qquad (229)$$

and so

$$\begin{aligned}
\left| \max_{0 \leq \ell \leq n-1} \max_{0 \leq j \leq n-1} |G(t_j, \theta_\ell)| - \max_{0 \leq \ell \leq n-1} I_\infty(\theta_\ell) \right| &\leq \max_{0 \leq \ell \leq n-1} \left| \max_{0 \leq j \leq n-1} |G(t_j, \theta_\ell)| - I_\infty(\theta_\ell) \right| \\
&\leq C L^2 \left\| \frac{\partial^2 G}{\partial t^2} \right\|_{L^\infty} \frac{1}{n^2}, \qquad (230)
\end{aligned}$$

as claimed.

□

Putting together Lemma 4.11 and Lemma 4.12, for all n sufficiently large,

$$\left| \left(\frac{2}{n} \sum_{\ell=0}^{n-1} \frac{L}{n} \sum_{j=0}^{n-1} |G(t_j, \theta_\ell)|^p \right)^{1/p} - \|f\|_{SV_p} \right| \leq C L^{2+1/p} \left(\left\| \frac{\partial^2 G}{\partial t^2} \right\|_{L^\infty} + \left\| \frac{\partial^2 G}{\partial \theta^2} \right\|_{L^\infty} \right) \frac{1}{n^2}. \qquad (231)$$

Combining this with (207), for all n sufficiently large

$$|\mathrm{SV}_p(x) - \|f\|_{SV^p}| \leq \frac{C}{n^2}, \qquad (232)$$

completing the proof of Proposition 4.7.

4.2.2 Proof of Proposition 4.8

First, we note that the third part of the Proposition (almost sure convergence) follows immediately from the second part (concentration bound) by using Lemma 6.7, as in the 1D case.

Recall that the vectors α_θ and β_θ are defined as follows:

$$\alpha_\theta[k] = \frac{L^2}{n^2} \sum_{i,j} Z[i,j] e^{-2\pi \mathrm{i} k (t_i \cos(\theta) + t_j \sin(\theta))/L}, \quad 0 < |k| < n/2, \qquad (233)$$

$$\beta_\theta[k] = \frac{\alpha_\theta[k]}{2\pi \mathrm{i} k/L}, \quad 0 < |k| < n/2, \qquad (234)$$

and when $k = 0$,

$$\beta_\theta[0] = - \sum_{0 < |k| < n/2} \beta_\theta[k](-1)^k. \qquad (235)$$

Define the random vector W by $W[j,\ell] = \nu_Z(t_j, \theta_\ell)$, that is,

$$W[j,\ell] = \frac{1}{L} \sum_{k=-n/2+1}^{n/2-1} \beta_{\theta_\ell}[k] e^{2\pi \mathrm{i} t_j k/L} \tag{236}$$

for $0 \le j, \ell \le n-1$. Then for each $1 \le p < \infty$,

$$\mathrm{SV}_p(Z_n) = \left(\frac{L}{n^2} \sum_{\ell=0}^{n-1} \sum_{j=0}^{n-1} |W[j,\ell]|^p \right)^{1/p}, \tag{237}$$

and $\mathrm{SV}_\infty(Z_n) = \|W\|_\infty$.

We define the auxiliary vector X by

$$X[j,\ell] = \sum_{0<|k|<n/2} \beta_{\theta_\ell}[k] e^{2\pi \mathrm{i} t_j k/L}. \tag{238}$$

Note that $W[j,\ell] = (X[j,\ell] + \beta_{\theta_\ell}[0])/L$.

Expectation of $\mathrm{SV}_p(Z_n)$**.** For fixed $0 \le i, \ell \le n-1$,

$$\begin{aligned} X[i,\ell] &= \sum_{0<|k|<n/2} \beta_{\theta_\ell}[k] e^{2\pi \mathrm{i} t_i k/L} \\ &= \frac{L}{2\pi \mathrm{i}} \sum_{0<|k|<n/2} \frac{\alpha_{\theta_\ell}[k]}{k} e^{2\pi \mathrm{i} t_i k/L} \\ &= \frac{L^3}{2\pi \mathrm{i} n^2} \sum_{0<|k|<n/2} \frac{1}{k} \sum_{j,j'} Z[j,j'] e^{-2\pi \mathrm{i} k (t_j \cos(\theta_\ell) + t_{j'} \sin(\theta_\ell))/L} e^{2\pi \mathrm{i} t_i k/L} \\ &= \frac{L^3}{\pi n^2} \sum_{j,j'} Z[j,j'] \sum_{k=1}^{n/2-1} \frac{1}{k} \sin(k 2\pi (t_j \cos(\theta_\ell) + t_{j'} \sin(\theta_\ell) - t_i)/L). \\ &= \frac{L^3}{\pi n^2} \sum_{j,j'} Z[j,j'] Q_n[i,j,j',\ell], \end{aligned} \tag{239}$$

where

$$Q_n[i,j,j',\ell] = \sum_{k=1}^{n/2-1} \frac{1}{k} \sin(k 2\pi (t_j \cos(\theta_\ell) + t_{j'} \sin(\theta_\ell) - t_i)/L). \tag{240}$$

From Lemma 6.6, $|Q_n[i,j,j',\ell]| \le C$ for all i, j, j' and ℓ and for a universal constant C. Since the $Z[j,j']$ are independent and mean zero, we may apply Proposition 6.11 to get the bound

$$\|X[i,\ell]\|_{\psi_2}^2 \le C \frac{L^6}{n^4} \sum_{j,j'} Q_n[i,j,j',\ell]^2 \|Z[j,j']\|_{\psi^2}^2 \le C L^6 \frac{\sigma^2}{n^2}. \tag{241}$$

A nearly identical proof works for $\beta[0]$, showing that

$$\|\beta[0]\|_{\psi_2}^2 \le C L^6 \frac{\sigma^2}{n^2}. \tag{242}$$

Since $W[j,\ell] = (X[j,\ell] + \beta_{\theta_\ell}[0])/L$, it then follows that

$$\|W[j,\ell]\|_{\psi_2} \le C L^2 \frac{\sigma}{n}. \tag{243}$$

and hence, for any $1 \le p < \infty$, from Proposition 6.8,

$$\mathbb{E}[|W[j,\ell]|^p] \le \left(C\sqrt{p}L^2\frac{\sigma}{n}\right)^p. \tag{244}$$

Consequently,

$$\mathbb{E}[\mathrm{SV}_p(Z_n)^p] = \frac{L}{n^2}\sum_{j,\ell}\mathbb{E}[|W[j,\ell]|^p] \le \left(C\sqrt{p}L^{2+1/p}\frac{\sigma}{n}\right)^p, \tag{245}$$

and therefore, using Jensen's inequality,

$$\mathbb{E}[\mathrm{SV}_p(Z_n)] \le (\mathbb{E}[\mathrm{SV}_p(Z_n)^p])^{1/p} \le C\sqrt{p}L^{2+1/p}\frac{\sigma}{n}. \tag{246}$$

The case when $p = \infty$ follows immediately from Lemma 6.10.

Concentration of $\mathrm{SV}_p(Z_n)$. Fix $1 \le p < \infty$, so that

$$\mathrm{SV}_p(Z_n)^p = \frac{L}{n^2}\sum_{j=0}^{n-1}\sum_{\ell=0}^{n-1}|W[j,\ell]|^p. \tag{247}$$

Since each $\|W[j,\ell]\|_{\psi_2} \le CL^2\sigma/n$, therefore, by Lemma 6.14, $\||W[j,\ell]|^p\|_{\psi_{2/p}} \le C^pL^{2p}\sigma^p/n^p$, and so by Proposition 6.13,

$$\|\mathrm{SV}_p(Z_n)^p\|_{\psi_{2/p}} \le C_pL^{2p+1}\sigma^p/n^p. \tag{248}$$

Therefore, $\|\mathrm{SV}_p(Z_n)\|_{\psi_2} \le C_pL^{2+1/p}\sigma/n$, and hence, for all $t \ge 0$,

$$\mathbb{P}\{\mathrm{SV}_p(Z_n) \ge t\} \le 2\exp\left\{-C_p\frac{n^2t^2}{L^{4+2/p}\sigma^2}\right\}, \tag{249}$$

as claimed.

When $p = \infty$, we use Proposition 6.11 to get

$$\|\mathrm{SV}_\infty(Z_n)\|_{\psi_2} \le C\sqrt{\log(n)}\max_{0\le j,\ell\le n-1}\|W[j,\ell]\|_{\psi_2} \le CL^2\sigma\frac{\sqrt{\log(n)}}{n}. \tag{250}$$

Consequently, for any $t \ge 0$,

$$\mathbb{P}\{\mathrm{SV}_\infty(Z_n) \ge t\} \le 2\exp\left\{-C\frac{n^2t^2}{L^4\sigma^2\log(n)}\right\}, \tag{251}$$

which is the claimed bound.

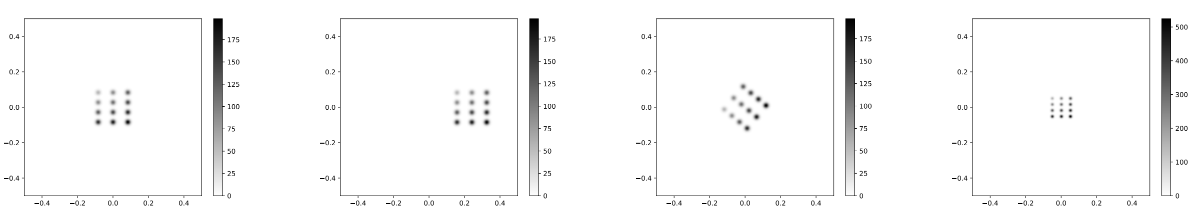

Figure 1: The function f described in Section 5.1, and examples of the deformations applied to f. From left to right: the original function; a translation; a rotation; a dilation.

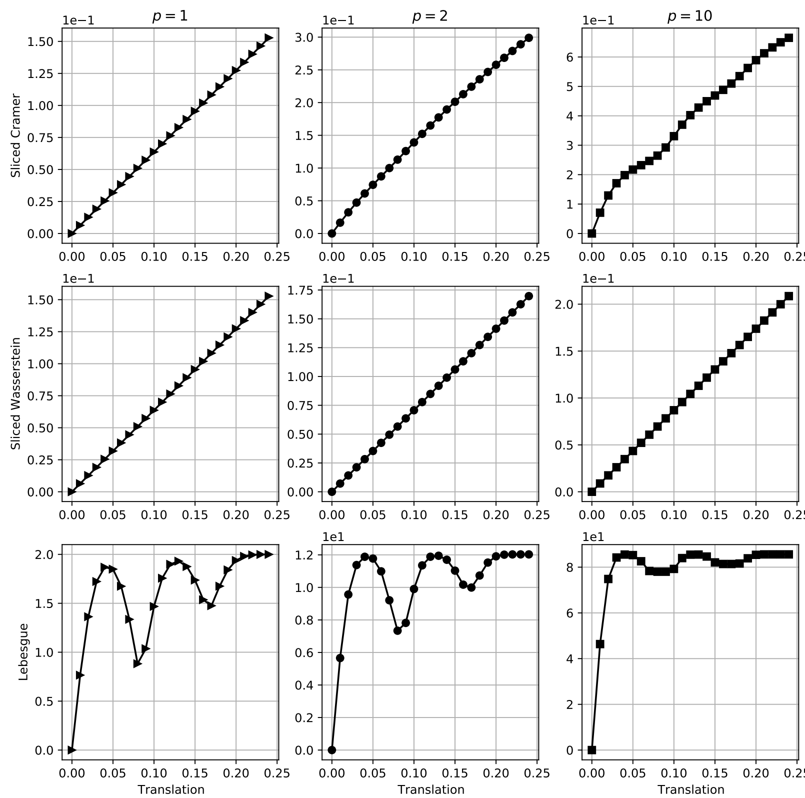

Figure 2: Results of the experiment described in Section 5.1, showing the distances from f to its translations as a function of the translation size.

5 Numerical results

In this section, we report on numerical experiments that explore the behavior of the sliced Cramér metrics in 2D under deformations and noise, illustrating their behavior on selected examples. We compare and contrast the behavior of the sliced Cramér metrics with that of the sliced Wasserstein and Lebesgue distances. Code for the sliced Cramér and sliced Wasserstein distances may be found at `https://github.com/wleeb/SlicedCramer`.

5.1 Deformations in 2D

We consider the function f shown in the leftmost panel of Figure 1, which is a convex combination of 12 isotropic Gaussian functions:

$$f(x) = \sum_{i=1}^{3} \sum_{j=1}^{4} w_{ij} e^{-|x-c_{ij}|^2/\tau}, \tag{252}$$

where $\tau = 1/5000$. Although f is, strictly speaking, not compactly supported, it is negligibly small outside of $[-1/2, 1/2] \times [-1/2, 1/2]$. The centers c_{ij} of the Gaussians comprising f are arranged in a 4-by-3 grid, equispaced in the rectangle $[-1/12, 1/12] \times [-1/12, 1/12]$. The weights w_{ij} assigned to the Gaussians in f may be described as follows: assigning the numbers 1 to 4 to the rows going from top to bottom, and assigning 1 to 3 to the columns going from left to right, the weight assigned to the Gaussian in position (i, j) is proportional to $\sqrt{i^2 + j^2}$. These weights are then normalized to sum to 1, so that $\|f\|_{L^1} = 1$.

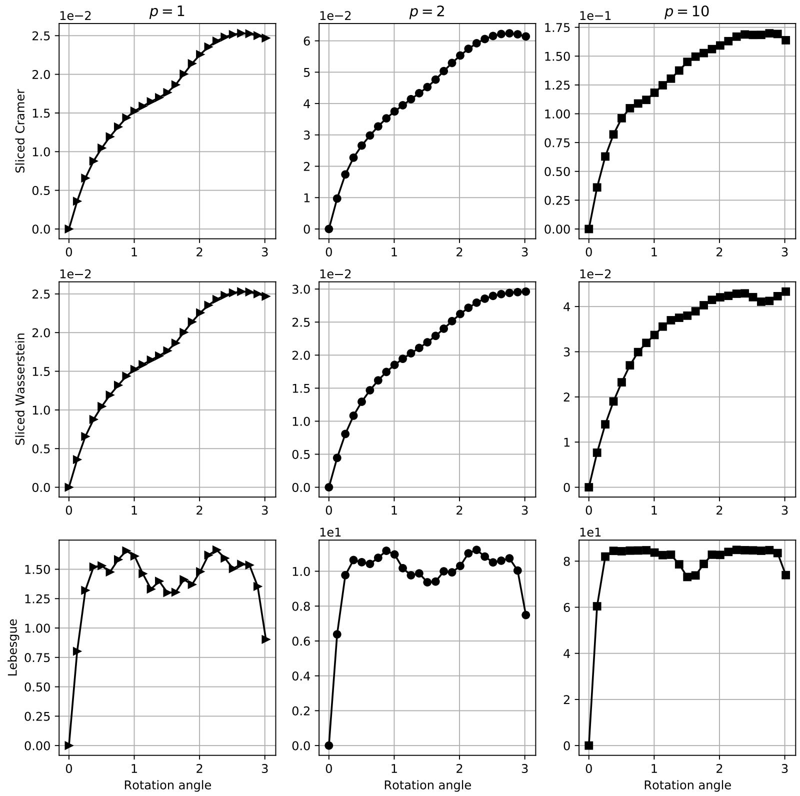

Figure 3: Results of the experiment described in Section 5.1, showing the distances from f to its rotations as a function of the rotation angle.

We examine the distances between f and its deformations. For each metric D, we compute the distances $\mathrm{D}(f, f_{\Phi_\delta})$, where Φ_δ is a deformation depending on a single parameter δ, where the displacement of Φ_δ grows with δ. The distances D are the sliced p-Cramér, sliced p-Wasserstein, and p-Lebesgue, for $p = 1, 2, 10$. We consider three types of deformations: translations, rotations, and dilations. Examples of these are displayed in Figure 1. In all examples, we evaluate the distances for 25 deformation parameters, using samples on a 500-by-500 grid.

Figure 2 shows the distances $\mathrm{D}(f, f_{\Phi_\delta})$ as a function of the translation size δ, where $\Phi_\delta(x, y) = (x + \delta, y)$. Figure 3 shows the distances $\mathrm{D}(f, f_{\Phi_\delta})$ as a function of the rotation angle δ, where $\Phi_\delta(x, y) = (x\cos(\delta) - y\sin(\delta), x\sin(\delta) + y\cos(\delta))$. Figure 4 shows the distances $\mathrm{D}(f, f_{\Phi_\delta})$ as a function of the dilation parameter $\delta \geq 1$, where $\Phi_\delta(x) = \delta x$.

From the plots, it is evident that the sliced p-Cramér and sliced p-Wasserstein distances exhibit similar behavior. Both metrics also change more smoothly than the Lebesgue distances as the deformation parameter changes, particularly for translation and rotation. For dilation, the Lebesgue 1-distance quickly becomes large and nearly constant, because the supports of f and its dilations become nearly disjoint as the dilation size grows. On the other hand, the Lebesgue p-distances for $p > 1$ appear to vary more smoothly with the dilation parameter; this is because the p-norm of the dilated function grows with the dilation size, and so the distance in this case is due to the growing size of the single function, rather than providing any meaningful information about the relationship between the two functions. By contrast, the sliced p-Cramér distance does not grow arbitrarily big as the norm grows; see Remark 8 after the statement of Theorem 3.1.

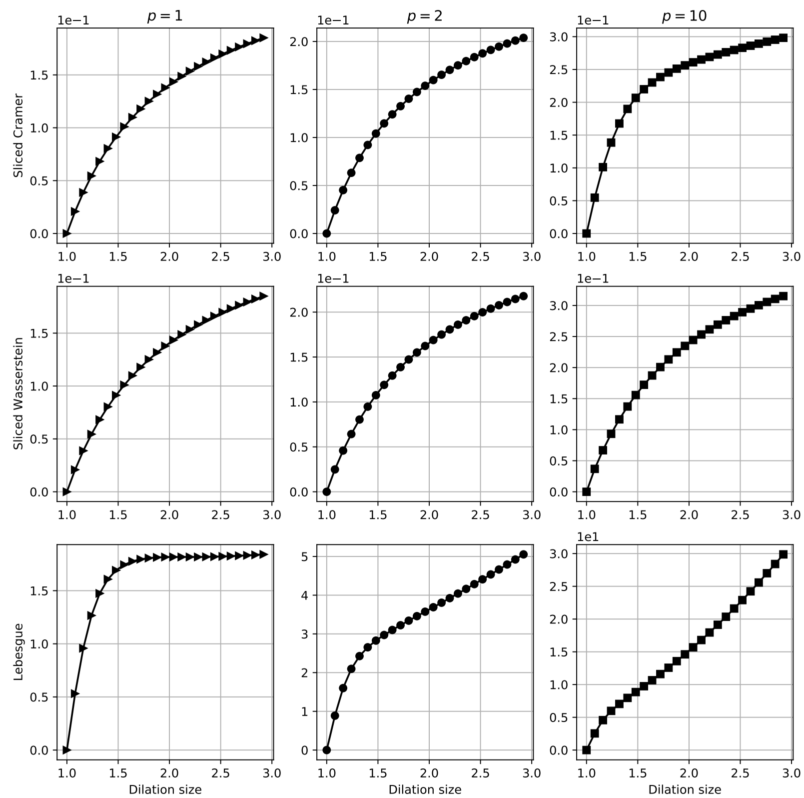

Figure 4: Results of the experiment described in Section 5.1, showing the distances from f to its dilations as a function of the dilation size.

5.2 Rotations and projections in 3D

We illustrate Corollary 3.2 by comparing the sliced Cramér metrics to the sliced Wasserstein and Lebesgue distances between 2D projections of rotated volumes. We consider the function f on $\mathbb{R}^3$ defined as a convex combination of 26 spherical Gaussians centered on a spiral circling the z-axis:

$$f(x) = \sum_{j=1}^{26} w_j e^{-|x-c_j|^2/\tau}, \tag{253}$$

where $\tau = 1/200$, and the centers are located at the points

$$c_j = (0.5\cos(4\pi j/26), 0.5\sin(4\pi j/26), -1/2 + j/25), \tag{254}$$

with respective heights w_j proportional to $1.5 + \cos(4\pi j/26)$, where $0 \leq j \leq 25$, normalized to sum to 1, so that $\|f\|_{L^1} = 1$. The function is rotated within the yz-plane, with rotations between 0 and $\pi/4$, and then analytically projected onto the xy-plane, sampled on a 500-by-500 grid over $[-r, r] \times [-r, r]$, with $r = 2.5$. Example projections are displayed in Figure 5. We evaluate the distances based on these samples as the rotation angle grows; the distances are displayed in Figure 6. It is evident that the sliced Cramér and sliced Wasserstein distances grow more slowly with the rotation angle, and exhibit smoother growth, than do the Lebesgue distances.

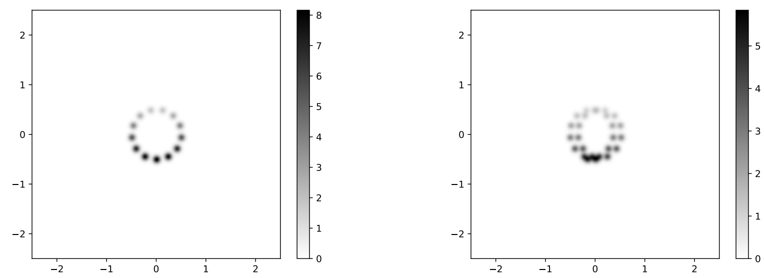

Figure 5: Projections used in the experiment described in Section 5.2. Left: the projection onto the xy-plane of the original spiral function of f. Right: the projection onto the xy-plane of a rotation of f within the yz-plane.

5.3 Robustness to noise

We examine the robustness of the sliced Cramér distances to additive noise. We consider the function f from Section 5.1, and add Gaussian noise to the samples of rotations of f on a grid of size $n = 512$. We then compute both the sliced Cramér distances and the Euclidean distances between f and the noisy samples of its rotations. The noise standard deviations are chosen to be $\sigma = 20, 30, 40$. The distances are averaged over 20 runs of the experiment. The resulting plots are shown in Figure 7. It is evident that the sliced Cramér distances are quite robust to noise at this sample size, in the sense that though the distances are inflated, they still closely track the distances between the noiseless functions (the curve where $\sigma = 0$). For reference, we also plot the Lebesgue distances, which exhibit much greater inflation due to noise.

For a more quantitative exploration of the effect of noise on the sliced Cramér distances, we show the following experiment. For increasing values of n, we sample the function f and a translate of f by $1/5$ to the right, which we denote by g, on a grid of size n-by-n, and add Gaussian noise with standard deviation 20 to the samples of g. For $p = 1, 2, 10$, we evaluate the sliced p-Cramér distances between f and the noisy samples of g. For each n, the experiment is repeated $M = 1000$ times. We estimate the true distance d between f and g by evaluating them on a grid of size 2048-by-2048, and measure the average absolute relative error between the noisy distances $\widehat{d}_k$ and d:

$$\mathrm{err}_{n,p} = \frac{1}{M} \sum_{k=1}^{M} \frac{|d - d_k|}{d}. \tag{255}$$

These value are plotted against n^2 in the right panel of Figure 8, in log scale. We also measure the average sliced Volterra p-norms of the noise itself, plotted against n^2 in the left panel of Figure 8. The average errors decay approximately like $O(1/n)$ as n increases (that is, the slopes of the plots are close to $-1/2$), consistent with the error rate established in Theorem 4.6.

6 Additional proof details

In this section, we state a number of results that are used throughout the proofs in the paper. Some of these results are either well-known or fairly straightforward, and we place them here to avoid interrupting the main text. We provide proofs of those results which we were unable to locate in the literature.

6.1 Fourier analysis and approximation theory

The following is a basic lemma on the decay and approximation of the Fourier transform for C^r functions:

Lemma 6.1. *Suppose $f : \mathbb{R}^d \to \mathbb{R}$ is C^r, where $r > d$, and is supported in $[0, L]^d$. Then $|\widehat{f}(\nu)| = O(|\nu|^{-r})$, and for any $\xi \in \mathbb{R}^d$ with $\|\xi\|_\infty \le n/(2L)$ and any $n \ge 2$,*

$$\left| \frac{1}{n^d} \sum_{k \in \{0,\dots,n-1\}^d} f(Lk/n) e^{-2\pi \mathrm{i} \langle Lk, \xi \rangle / n} - \widehat{f}(\xi) \right| = O(n^{-r}). \tag{256}$$

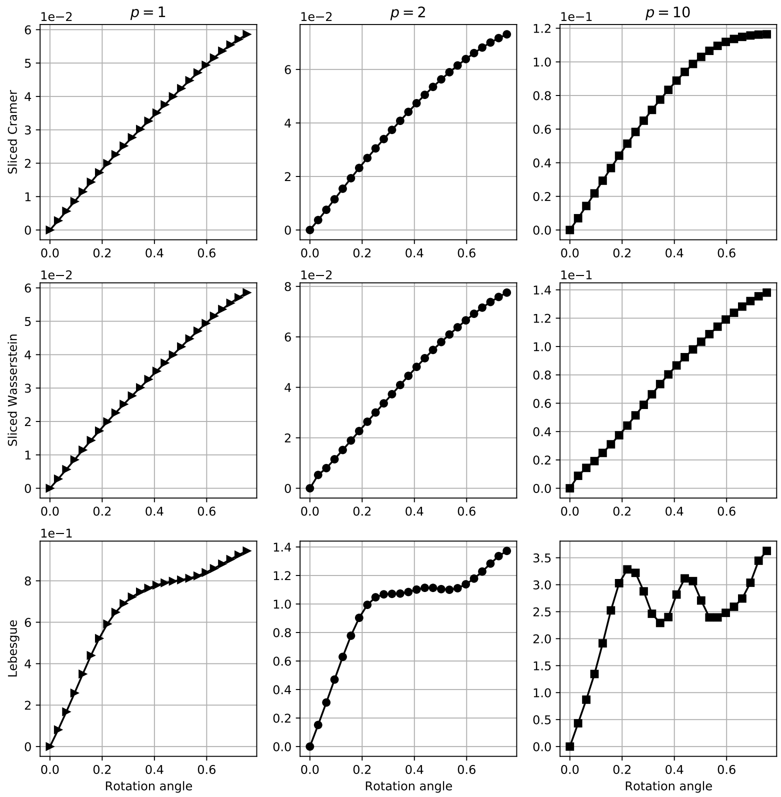

Figure 6: Results of the experiment described in Section 5.2, showing the distances from the projection of f to the projections of its rotation as a function of the rotation angle.

We will also make use of a special case of the periodic version:

Lemma 6.2. *Suppose $f : \mathbb{R} \to \mathbb{R}$ is L-periodic, continuous, and has Fourier series*

$$f(x) = \frac{1}{L} \sum_{k=-\infty}^{\infty} c_k e^{2\pi \mathrm{i} kx/L} \tag{257}$$

satisfying $|c_k| \leq M/|k|^r$ for $r > 1$ and a constant $M > 0$. Then there is a value $C = C(r) > 0$ such that for all $n \geq 2$,

$$\left| \frac{L}{n} \sum_{j=0}^{n-1} f(jL/n) - \int_0^L f(x)dx \right| \leq C \frac{M}{n^r}, \tag{258}$$

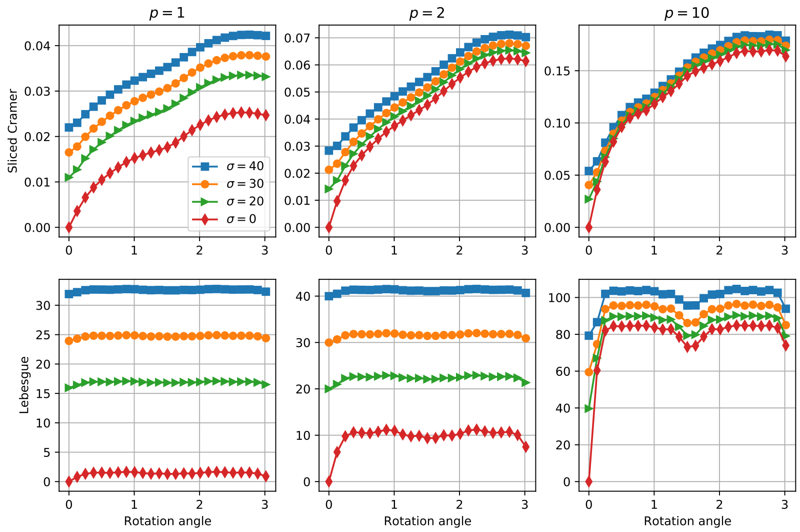

Figure 7: Results of the experiment described in Section 5.3, showing the distances between f and its noisy rotations as the rotation angle increases and for different noise levels.

Proof. We have:

$$\begin{aligned}\frac{L}{n}\sum_{j=0}^{n-1} f(jL/n) &= \frac{1}{n}\sum_{j=0}^{n-1}\sum_{k\in\mathbb{Z}} c_k e^{2\pi \mathrm{i} kj/n} \\ &= \int_0^L f(x)dx + \sum_{k\neq 0} c_k \left(\frac{1}{n}\sum_{j=0}^{n-1} e^{2\pi \mathrm{i} kj/n}\right) \\ &= \int_0^L f(x)dx + \sum_{k=0 \bmod n,\, k\neq 0} c_k, \end{aligned} \tag{259}$$

and consequently

$$\left|\frac{L}{n}\sum_{j=0}^{n-1} f(t_j) - \int_0^L f(x)dx\right| \le \sum_{\ell\neq 0}|c_{\ell n}| \le M\sum_{\ell\neq 0}\frac{1}{|\ell n|^r} = \frac{M}{n^r}\left(2\sum_{\ell=1}^{\infty}\ell^{-r}\right). \tag{260}$$

□

Next, we state several results that bound the error on the approximation the L^p norm of a periodic function using the trapezoidal rule.

Lemma 6.3. *Suppose $G : \mathbb{R} \to \mathbb{R}$ is L-periodic and C^2. Then for any $1 \le p < \infty$ and integer $k \neq 0$,*

$$\left|\int_0^L |G(t)|^p e^{-2\pi \mathrm{i} kt/L}\, dt\right| \le C\frac{L^2 p}{k^2}\|G''\|_{L^\infty}\int_0^L |G(t)|^{p-1}dt, \tag{261}$$

where $C > 0$ is a universal constant.

Proof. First, if $\mathrm{sign}(G(t))$ is constant, then the function $|G(t)|^p$ is C^2, and the proof we give below can be simplified to give the desired bound, using integration-by-parts twice; we will skip the details of this case. We will therefore assume that G has at least one root, but is also not constantly zero. In this case, because G is L-periodic, we can assume without loss of generality that $G(0) = G(L) = 0$.

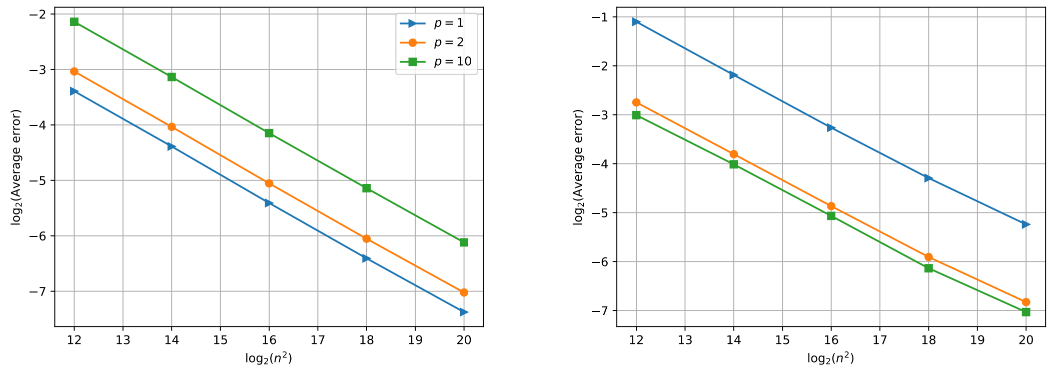

Figure 8: Results of the experiment described in Section 5.3, showing the average errors as a function of number of samples (in log scale). Left: sliced Volterra norms of noise alone. Right: average relative errors between the noisy distances and the true distances.

Let $A = \{t \in (0, L) : G(t) \neq 0\}$. This set is open and non-empty, hence can be written as the disjoint union of at most countably many disjoint intervals (a_j, b_j). Note that $\operatorname{sign}(G(t))$ is constant for $t \in (a_j, b_j)$. Also, note that $G(a_j) = G(b_j) = 0$ for all j; indeed, if $a_j = 0$, then $G(a_j) = 0$ by assumption; and if $a_j > 0$ but $G(a_j) \neq 0$, then there would be some j' with $a_j \in (a_{j'}, b_{j'}) \subset A$; but this contradicts that the intervals (a_j, b_j) and $(a_{j'}, b_{j'})$ are disjoint.

Let $H_p(x) = |G(x)|^p$. Suppose first that $p > 1$. Then on each interval (a_j, b_j),

$$H_p'(x) = p \operatorname{sign}(G(x))|G(x)|^{p-1}G'(x), \tag{262}$$

and

$$H_p''(x) = p(p-1)|G(x)|^{p-2}G'(x)^2 + p \operatorname{sign}(G(x))|G(x)|^{p-1}G''(x). \tag{263}$$

Since $H_p(0) = H_p(L)$, we have, for all $k \neq 0$,

$$\int_0^L H_p(x)e^{-2\pi \mathrm{i}kx/L}\,dx = \frac{L}{2\pi \mathrm{i}k}\int_0^L H_p'(x)e^{-2\pi \mathrm{i}kx/L}\,dx. \tag{264}$$

Let $\chi(x, j)$ be 1 if $x \in (a_j, b_j)$, and 0 otherwise. We then write

$$\int_0^L H_p'(x)e^{-2\pi \mathrm{i}kx/L}\,dx = \int_0^L \sum_j \chi(x, j)H_p'(x)e^{-2\pi \mathrm{i}kx/L}\,dx, \tag{265}$$

and since

$$\leq \int_0^L \sum_j \chi(x, j)|H_p'(x)|\,dx \leq L\|H_p'\|_{L^\infty} < \infty, \tag{266}$$

we may use Fubini's Theorem to switch the order of summation and integration to obtain

$$\begin{aligned}\int_0^L H_p'(x)e^{-2\pi \mathrm{i}kx/L}\,dx &= \int_0^L \sum_j \chi(x, j)H_p'(x)e^{-2\pi \mathrm{i}kx/L}\,dx \\ &= \sum_j \int_0^L \chi(x, j)H_p'(x)e^{-2\pi \mathrm{i}kx/L}\,dx \\ &= \sum_j \int_{a_j}^{b_j} H_p'(x)e^{-2\pi \mathrm{i}kx/L}\,dx. \end{aligned} \tag{267}$$

We now bound this sum. Using integration-by-parts again gives

$$\begin{aligned}\int_{a_j}^{b_j} H_p'(x)e^{-2\pi \mathrm{i}kx/L}\,dx &= \frac{H_p'(a_j^+)e^{-2\pi\mathrm{i}ka_j/L} - H_p'(b_j^-)e^{-2\pi\mathrm{i}kb_j/L}}{2\pi\mathrm{i}k/L} + \frac{L}{2\pi\mathrm{i}k}\int_{a_j}^{b_j} H_p''(x)e^{-2\pi\mathrm{i}kx/L}\,dx \\ &= \frac{L}{2\pi\mathrm{i}k}\int_{a_j}^{b_j} H_p''(x)e^{-2\pi\mathrm{i}kx/L}\,dx, \end{aligned} \tag{268}$$

where the end terms vanish because $G(a_j) = G(b_j) = 0$ and $H_p'(x) = p\,\mathrm{sign}(G(x))|G(x)|^{p-1}G'(x)$. Now, suppose without loss of generality that $G(x) > 0$ on (a_j, b_j) (an analogous argument will work if $G(x) < 0$). Then

$$\left|\int_{a_j}^{b_j} H_p''(x)e^{-2\pi\mathrm{i}kx/L}\,dx\right| \le p\int_{a_j}^{b_j}(p-1)G(x)^{p-2}G'(x)^2dx + p\int_{a_j}^{b_j} G(x)^{p-1}|G''(x)|dx. \tag{269}$$

For the first integral, we let $D(x) = G(x)^{p-1}$, so that $D'(x) = (p-1)G(x)^{p-2}G'(x)$; then using integration-by-parts,

$$\begin{aligned} p\int_{a_j}^{b_j}(p-1)G(x)^{p-2}G'(x)^2dx &= p\int_{a_j}^{b_j} D'(x)G'(x)dx \\ &= p(D(b_j)G'(b_j) - D(a_j)G'(a_j)) - p\int_{a_j}^{b_j} D(x)G''(x)dx \\ &= -p\int_{a_j}^{b_j} G(x)^{p-1}G''(x)dx, \end{aligned} \tag{270}$$

and therefore,

$$\left|\int_{a_j}^{b_j} H_p''(x)e^{-2\pi\mathrm{i}kx/L}\,dx\right| \le 2p\int_{a_j}^{b_j}|G(x)|^{p-1}|G''(x)|dx \le 2p\|G''\|_{L^\infty}\int_{a_j}^{b_j}|G(x)|^{p-1}dx. \tag{271}$$

Summing over j then gives

$$\begin{aligned}\left|\int_0^L H_p'(x)e^{-2\pi\mathrm{i}kx/L}\,dx\right| &= \left|\sum_j\int_{a_j}^{b_j} H_p'(x)e^{-2\pi\mathrm{i}kx/L}\,dx\right| \\ &= \left|\sum_j \frac{L}{2\pi\mathrm{i}k}\int_{a_j}^{b_j} H_p''(x)e^{-2\pi\mathrm{i}kx/L}\,dx\right| \\ &\le \frac{L}{2\pi|k|}\sum_j\left|\int_{a_j}^{b_j} H_p''(x)e^{-2\pi\mathrm{i}kx/L}\,dx\right| \\ &\le \frac{L}{2\pi|k|}\sum_j 2p\|G''\|_{L^\infty}\int_{a_j}^{b_j}|G(x)|^{p-1}dx \\ &\le \frac{Lp}{\pi|k|}\|G''\|_{L^\infty}\int_0^L|G(x)|^{p-1}dx. \end{aligned} \tag{272}$$

Finally, using (264), we have

$$\begin{aligned}\left|\int_0^L |G(x)|^p\,e^{-2\pi\mathrm{i}kx/L}\,dx\right| &= \left|\int_0^L H_p(x)e^{-2\pi\mathrm{i}kx/L}\,dx\right| \\ &= \frac{L}{2\pi|k|}\left|\int_0^L H_p'(x)e^{-2\pi\mathrm{i}kx/L}\,dx\right| \\ &\le \frac{L^2p}{2\pi^2k^2}\|G''\|_{L^\infty}\int_0^L|G(x)|^{p-1}dx. \end{aligned} \tag{273}$$

Taking the limit as $p \to 1^+$ also shows the corresponding bound for $p = 1$ as well:

$$\left| \int_0^L |G(x)| e^{-2\pi i k x/L} \, dx \right| \le \frac{L^3}{2\pi^2 k^2} \|G''\|_{L^\infty}, \tag{274}$$

which completes the proof.

□

Corollary 6.4. *Suppose $G : \mathbb{R} \to \mathbb{R}$ is L-periodic and C^2. Let*

$$t_j = \frac{j}{n} L, \quad 0 \le j \le n. \tag{275}$$

Then for any $1 \le p < \infty$, and all $n \ge 2$,

$$\left| \frac{L}{n} \sum_{j=0}^{n-1} |G(t_j)|^p - \int_0^L |G(t)|^p dt \right| \le C \frac{L^2 p}{n^2} \|G''\|_{L^\infty} \int_0^L |G(t)|^{p-1} dt, \tag{276}$$

where C is a universal constant.

Proof. This follows from Lemma 6.2 and Lemma 6.3.

□

Corollary 6.5. *Suppose $G : \mathbb{R} \to \mathbb{R}$ is L-periodic and C^2. Let*

$$t_j = \frac{j}{n} L, \quad 0 \le j \le n. \tag{277}$$

Then for any $1 \le p < \infty$, there is N_p such that for all $n \ge N_p$,

$$\left| \left(\frac{L}{n} \sum_{k=0}^{n-1} |G(t_k)|^p \right)^{1/p} - \left(\int_0^L |G(t)|^p dt \right)^{1/p} \right| \le C \frac{L^{2+1/p} \|G''\|_{L^\infty}}{n^2}, \tag{278}$$

with the obvious modification when $p = \infty$, where C is a universal constant.

Proof. Suppose G is not constantly 0 (otherwise the result is trivial). For brevity, let $I = \|G\|_{L^p}^p / 2$. Let

$$T_{n,p} = \frac{L}{n} \sum_{k=0}^{n-1} |G(t_k)|^p. \tag{279}$$

Then for all n sufficiently large, $T_{n,p} \ge I$. The function $y \mapsto y^{1/p}$ has derivative $y^{1/p-1}/p$, which has maximum value $I^{1/p-1}/p$ when $y \ge I$. Hence, by the mean value theorem, for all n sufficiently large,

$$\begin{aligned}
\left| T_{n,p}^{1/p} - \|G\|_{L^p} \right| &\le |T_{n,p} - \|G\|_{L^p}^p| \, (I^{1/p-1}/p) \\
&\le C L^{2+1/p} p \|G\|_{L^p}^{p-1} \|G''\|_{L^\infty} \frac{1}{n^2} \frac{(\|G\|_{L^p}^p/2)^{1/p-1}}{p} \\
&\le C \frac{L^{2+1/p} \|G''\|_{L^\infty}}{n^2}.
\end{aligned} \tag{280}$$

This is the desired bound for $1 \le p < \infty$.

When $p = \infty$, suppose again that G is not constantly zero, since otherwise the result is trivial. Let x^* satisfy $\|G\|_{L^\infty} = |G(x^*)|$. If $x^* = 0$, then then, since $t_0 = 0$, $\max_{0 \le k \le n-1} |G(t_k)| = |G(0)| = \|G\|_{L^\infty}$. (If $x^* = L$, then since G is L-periodic, we can also take $x^* = 0$.) Assume, then, that x^* lies in the interior of $(0, L)$. Then $G'(x^*) = 0$, and so a second-order Taylor expansion gives

$$|G(x) - G(x^*)| \le C \|G''\|_{L^\infty} |x - x^*|^2, \tag{281}$$

where $C > 0$ is universal. Consequently, if t_{k^*} is a grid point within L/n of x^*, we have

$$|G(t_{k^*}) - G(x^*)| \leq C\frac{L^2}{n^2}\|G''\|_{L^\infty}, \tag{282}$$

and therefore, since $\max_{0\leq j\leq n-1}|G(t_j)| \geq |G(t_{k^*})|$,

$$\begin{aligned}\left|\max_{0\leq j\leq n-1}|G(t_j)| - \|G\|_{L^\infty}\right| &= |G(x^*)| - \max_{0\leq j\leq n-1}|G(t_j)| \\ &\leq |G(x^*)| - |G(t_{k^*})| \\ &\leq |G(x^*) - G(t_{k^*})| \\ &\leq C\frac{L^2\|G''\|_{L^\infty}}{n^2},\end{aligned} \tag{283}$$

which is the desired result.

□

The following result may be found (in more general form) in Chapter V of [86]:

Lemma 6.6. *There is a constant $C > 0$ such that for any positive integer m and real number A,*

$$\left|\sum_{k=1}^{m}\frac{\sin(kA)}{k}\right| \leq C. \tag{284}$$

6.2 Probability and concentration

We start with the following well-known corollary of the Borel-Cantelli Lemma (see e.g. Chapter 2, Section 10 of [67]):

Lemma 6.7. *Let $R_1, R_2, \ldots$ be a sequence of random numbers. Suppose that for all $\epsilon > 0$,*

$$\sum_{n=1}^{\infty}\mathbb{P}\{R_n > \epsilon\} < \infty. \tag{285}$$

Then $R_n \to 0$ almost surely.

6.2.1 Properties of sub-Gaussian random variables

Proposition 6.8. *If R is a sub-Gaussian random variable, then for any $p \geq 1$,*

$$(\mathbb{E}[|R|^p])^{1/p} \leq C\sqrt{p}\|R\|_{\psi_2}, \tag{286}$$

where $C > 0$ is a universal constant.

Proposition 6.9. *If R is a sub-Gaussian random variable with mean zero, then for any $s \in \mathbb{R}$,*

$$\mathbb{E}[e^{sR}] \leq e^{Cs^2\|R\|_{\psi_2}^2}, \tag{287}$$

where $C > 0$ is a universal constant.

The proofs of Proposition 6.8 and Proposition 6.9 may be found in [75]. Next, we prove a standard bound on the expected value of the maximum of sub-Gaussians:

Lemma 6.10. *Let $R[0], R[1], \ldots, R[n-1]$ be mean zero sub-Gaussian random variables, with sub-Gaussian norms $\|R[j]\|_{\psi_2} \leq \tau$, $0 \leq j \leq n-1$, and let $R = (R[0], \ldots, R[n-1])$. Then*

$$\mathbb{E}[\|R\|_\infty] \leq C\tau\sqrt{\log(n)}, \tag{288}$$

where $C > 0$ is a universal constant.

Proof. First, extend R by defining $R[j] = -R[j-n]$ for $n \leq j \leq 2n-1$. Then

$$\begin{aligned} \|R\|_\infty &= \max\{R[0], \ldots, R[n-1], -R[0], \ldots, -R[n-1]\} \\ &= \max\{R[0], \ldots, R[n-1], R[n], \ldots, R[2n-1]\}. \end{aligned} \tag{289}$$

Note that for all j, since $R[j]$ is a mean-zero sub-Gaussian with $\|R[j]\|_{\psi_2} \leq \tau$, by Proposition 6.9 the moment generating functions satisfy the bound

$$\mathbb{E}[e^{sR[j]}] \leq e^{C\tau^2 s^2}. \tag{290}$$

Using this bound, for any $s > 0$, we have

$$\begin{aligned} e^{\mathbb{E}[s\|R\|_\infty]} &\leq \mathbb{E}\left[e^{s\|R\|_\infty}\right] \\ &= \mathbb{E}\left[\max_{0\leq j\leq 2n-1} e^{sR[j]}\right] \\ &\leq \mathbb{E}\left[\sum_{j=0}^{2n-1} e^{sR[j]}\right] \\ &= \sum_{j=0}^{2n-1} \mathbb{E}\left[e^{sR[j]}\right] \\ &\leq 2n \max_{0\leq j\leq 2n-1} \mathbb{E}\left[e^{sR[j]}\right] \\ &\leq 2ne^{C\tau^2 s^2} \end{aligned} \tag{291}$$

and so, taking the log of each side gives

$$\mathbb{E}\left[\|R\|_\infty\right] \leq C\left(\frac{\log(n)}{s} + s\tau^2\right), \tag{292}$$

and taking $s = \sqrt{\log(n)}/\tau$ gives the bound

$$\mathbb{E}\left[\|R\|_\infty\right] \leq C\tau\sqrt{\log(n)}, \tag{293}$$

as desired.

□

The next result bounds the sub-Gaussian norm of the maximum of sub-Gaussians; see Proposition 2.7.6 in [75].

Proposition 6.11. *If $R_1, \ldots, R_m$ are sub-Gaussian random variables, then*

$$\left\|\max_{1\leq i\leq m} R_i\right\|_{\psi_2} \leq C\sqrt{\log(m)} \max_{1\leq i\leq m} \|R_i\|_{\psi_2}, \tag{294}$$

where $C > 0$ is a universal constant.

The next result appears as Proposition 2.7.1 in [75].

Proposition 6.12. *There is a universal constant $C > 0$ such that if $R[0], \ldots, R[n-1]$ are independent sub-Gaussian random variables,*

$$\left\|\sum_{j=0}^{n-1} R[j]\right\|_{\psi_2}^2 \leq C\sum_{j=0}^{n-1} \|R[j]\|_{\psi_2}^2. \tag{295}$$

6.2.2 Sub-Weibull random variables

For $\alpha > 0$, define the function $\psi_\alpha(x) = \exp(x^\alpha) - 1$ on $[0, \infty)$. Note that ψ_α is increasing, and $\psi_\alpha(0) = 0$. Define

$$\|X\|_{\psi_\alpha} = \inf\{t > 0 : \mathbb{E}[\psi_\alpha(|X|/t)] \le 1\}. \tag{296}$$

Random variables X with $\|X\|_{\psi_\alpha} < \infty$ are called *sub-Weibull(α)*; see [33]. Note that $\|X\|_{\psi_\alpha}$ is a norm when $\alpha \ge 1$, since ψ_α is convex. When $0 < \alpha < 1$, $\|X\|_{\psi_\alpha}$ does not satisfy $\|X + Y\|_{\psi_\alpha} \le \|X\|_{\psi_\alpha} + \|Y\|_{\psi_\alpha}$, but the following still holds:

Proposition 6.13. *Let $\alpha > 0$. Then there is a value $C_\alpha > 0$ such that for all sub-Weibull(α) random variables $X_1, \dots, X_n$,*

$$\|X_1 + \cdots + X_n\|_{\psi_\alpha} \le C_\alpha(\|X_1\|_{\psi_\alpha} + \cdots + \|X_n\|_{\psi_\alpha}), \tag{297}$$

Proposition 6.13 is easily shown by adapting the proof of Proposition 2.6.1 in [75]. The following result is immediate from the definition of sub-Weibull:

Lemma 6.14. *Let $p \ge 1$. A random variable R is sub-Gaussian if and only if $X = |R|^p$ is sub-Weibull($2/p$), in which case $\|X\|_{\psi_{2/p}} = \|R\|_{\psi_2}^p$.*

7 Conclusion

This paper has proven a number of properties of sliced Cramér distances. We have characterized their growth under deformations of the inputs, proven stability to convolutions, provided bounds on the discretization errors, and shown robustness to heteroscedastic sub-Gaussian noise.

Looking forward, there are several questions that follow from the present work. First, at the core of the sliced Cramér distances are, of course, the 1D Cramér distances. These are quite simple objects, defined by applying a smoothing filter to the input functions and then evaluating the ordinary Lebesgue distance. It is therefore of interest to consider metrics based on other families of smoothing filters, and to characterize their behavior under deformations and robustness to noise. Such metrics are abundant in applications in both Euclidean and non-Euclidean settings, and, unlike the sliced Cramér distances, often involve filtering/smoothing the inputs at multiple scales [46, 45, 38, 37, 66, 44]. In future work, we plan to study such metrics' growth under deformations and their stability to noise.

Second, while this paper has shown theoretical properties of the sliced Cramér distances, it is of interest to explore their behavior in scientific applications, such as clustering images or volumes. For instance, Corollary 3.2 suggests that they may be well-suited for analyzing data from cryo-electron microscopy (cryo-EM), in which one observes two-variable projections of a three-variable volume (a molecule), at unknown viewing directions, from which the volume is to be determined [68, 7, 15]. It would be of interest to see whether sliced Cramér metrics perform well relative to other metrics that have been proposed for image clustering and parameterizing volumes in cryo-EM, such as Wasserstein-type metrics [70, 58, 79, 65]. One such application is heterogeneity analysis [82, 20, 41, 2, 35, 64, 73, 36], for which the use of Wasserstein distances has been proposed [58].

Finally, in certain applications one seeks metrics that are not only robust to all deformations, but invariant to a specific class of deformations, such as rotations and/or translations [65, 58, 81]. In principle, any metric can be made invariant to a specified set of deformations by simply minimizing the distance over the class of deformations. In some cases of practical interest, such as rigid alignment, this minimization can be done with only small extra computational cost [65, 57]. In the context of the present work, this raises interesting questions. First, it is known that in the presence of noise, alignment accuracy deteriorates [61, 1]; studying how invariant distances behave under noise, or more generally in any setting where alignment cannot be done to high precision, is therefore of interest. Second, it is natural to ask how introducing invariance to one class of deformations, such as rotations, impacts robustness to other deformations. Questions along these lines will be pursued in future work.

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